PakEmis Sector Technical Note TN01

A Reproducible Agricultural Greenhouse-Gas and Air-Pollutant Inventory for Pakistan with Mass-Conserving Spatial Allocation

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Abstract

Pakistan's agricultural emissions remain poorly resolved spatially, and the country lacks a gridded agricultural air-pollutant inventory based on national activity data. We develop a reproducible framework that calculates greenhouse-gas and air-pollutant emissions from a shared set of activity data and a common nitrogen accounting system. The framework combines official Pakistani statistics, open global datasets, and published IPCC and EMEP/EEA methods.

We construct a *hybrid Tier 1/Tier 2* inventory covering six IPCC agricultural categories and allocate the resulting emissions to provinces, a 0.05° grid, and 116 district units plus one residual unit. Activity data and emission factors determine the magnitude of each emission stream, while process-matched spatial proxies determine its distribution. Each spatial product is rescaled to its control total, ensuring exact reconciliation across national, provincial, gridded, and district estimates.

For 2024, we estimate agricultural greenhouse-gas emissions of 223 Mt CO_2e using AR5 GWP-100 values, with an analytical uncertainty of ±19% and a Monte-Carlo 95% interval of 185–271 Mt CO_2e. We also estimate 2.7 Mt of NH_3 and, from crop-residue burning, 0.65 Mt of CO and 54 kt of $PM_{2.5}$. The air-pollutant inventory is provided on the same grid and district geography and as a 1990–2024 national time series.

We evaluate district patterns against independent government crop records and compare the gridded ammonia surface with an independent 0.1° inventory for South Asia. The two ammonia inventories show strong spatial agreement, with a Pearson correlation of 0.90 and 78 % overlap among top-decile cells. However, our national estimate is approximately 1.8 times higher, with a broadly consistent ratio across provinces. Stream-level comparisons show that aggregate statistics conceal important differences among livestock, fertiliser, and residue-burning emissions.

The ammonia estimate is particularly sensitive to the assumed distribution of manure-management systems, which has not been measured nationally. Across plausible configurations, national NH_3 emissions range from 1.7 to 3.0 Mt. We therefore report structural and parametric uncertainty separately and treat the livestock-ammonia estimate as provisional. The open-source framework provides a reproducible basis for biennial transparency reporting and a nationally grounded gridded agricultural emissions inventory for Pakistan.

The sector described here is the largest component of a ten-sector national inventory built on the same engine, in which agriculture is 50.2 % of 444.96 Mt CO_2e (Ahmad, 2026). Two of its boundaries are accounted for against another sector rather than described in prose: dung-cake nitrogen leaves the manure-management balance here, and the part the household survey observes burned arrives as household fuel in the residential sector while the remainder is carried as an unresolved row outside the total; and the crop residue burned as household fuel (9.27 Mt dry matter, 122 192 TJ) leaves the residue balance here and arrives in the same fuel ledger—an exact transfer rather than a pair of independent estimates. Readers concerned with those transfers, with the cross-sector uncertainty treatment, or with where agriculture sits among national key categories should read the two papers together.

Keywords: agricultural greenhouse-gas inventory, air-pollutant emissions, ammonia, hybrid IPCC Tier 1/2, dasymetric mapping, gridded livestock, mass-balance reconciliation, district downscaling, Monte-Carlo uncertainty, Pakistan, MRV, transparency reporting

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PakEmis Sector Technical Note TN01. Technical documentation of one sector module of the PakEmis release identified below, deposited with the data as part of the documentation record. The national data-description article (`paper_national/`) is the only document that states the national totals as its own result.

Release provenance. This report documents one release of PakEmis and is generated against it. Where a sector report and the release's generated product catalogue (`outputs/pakemis_product_catalog.csv`) disagree, the catalogue is authoritative: sector reports are revised on their own cadence and may lag the products.

Release version	`1.0.0rc67`
Commit	`1876f398f85c`
Reporting period	FY2023-24 (2023-07-01 to 2024-06-30)
Baseline configuration	S (survey basis)
National total	444.96 Mt CO_2e (AR5) under S, the reference; 450.20 Mt under L
Release manifest	`13462fb4c2d1ea62` (SHA-256 of `outputs/release_manifest.json`, first 16)
Source manifest	`c1be6d314cfce1ea` (SHA-256 of `data/manifests/source_manifest.csv`, first 16)
Data-description article	`paper_national/` (ESSD, in preparation), same release

1. Introduction

National greenhouse-gas inventories provide the accounting basis for reporting under the Paris Agreement. Under the Enhanced Transparency Framework, countries regularly report sectoral emissions using Common Reporting Tables. In many developing countries, agriculture, forestry and other land use (AFOLU) accounts for a substantial share of national emissions but is also among the sectors with the weakest activity data.

Pakistan illustrates this problem clearly. Agriculture is one of the country's largest emitting sectors, yet its national inventory provides limited spatial detail. Some livestock estimates also continue to rely partly on projections from older censuses. National totals are necessary for international reporting, but they provide little information about where emissions occur or where mitigation and improved data collection should be targeted.

Agricultural activity also produces pollutants that affect air quality. Ammonia released from fertiliser and manure is the principal alkaline precursor of secondary inorganic particulate matter. Agricultural NH_3 can therefore contribute to $PM_{2.5}$ concentrations far from the fields and livestock systems where it originates. Crop-residue burning in the Punjab rice–wheat belt also releases primary $PM_{2.5}$, carbon monoxide and ozone precursors during the post-monsoon period, when particulate pollution in Lahore is particularly severe.

Despite this importance, Pakistan has no gridded agricultural air-pollutant inventory derived from national activity data. Air-quality modelling instead relies largely on global emissions products. These products provide valuable regional coverage, but their activity data and spatial assumptions cannot reliably distinguish, for example, an intensively cultivated canal-belt district from an extensive rangeland district.

A useful national system must therefore meet three requirements. First, it must be reproducible and updateable as new official statistics become available. Second, it must provide provincial, district and gridded estimates that support subnational planning and air-quality modelling. Third, its greenhouse-gas and air-pollutant calculations must use consistent activity data and nitrogen flows.

The third requirement is physically important. A unit of manure nitrogen may be lost as ammonia during housing, storage, grazing or field application, or it may remain in the system and contribute to direct or indirect nitrous-oxide emissions. Calculating these pathways in separate workbooks creates a risk of double counting, omission, or inconsistent system boundaries. Similar inconsistencies arise when a national inventory, a spatial-allocation exercise, and an emissions input for a chemistry model are developed independently. The resulting products may appear detailed while failing to reproduce the same national total or physical flow.

The same argument applies one level up. An agricultural inventory that is internally consistent may still be inconsistent with the sectors around it, because several quantities are genuinely shared: nitrogen leaving the manure-management balance as household fuel has to arrive somewhere, and crop-residue burning has to be distinguishable from other open burning. The framework described here is the agriculture module of a ten-sector national inventory (Ahmad, 2026), in which agriculture is 50.2 % of the national total and those transfers are executable assertions rather than statements in a methods document. This paper documents the sector; the companion documents what holds between sectors, and neither repeats the other.

Many of the required inputs already exist. Pakistan publishes national and subnational agricultural statistics, while open global datasets provide spatial information on livestock, crops, land cover and fire occurrence. The remaining need is an integrated framework that calculates emissions, allocates them spatially under an explicit mass-balance constraint, evaluates the resulting products, and propagates uncertainty from a common set of parameters.

This study develops such a framework for Pakistan. We construct a *hybrid Tier 1/Tier 2* agricultural inventory using the 2006 IPCC Guidelines and the 2019 Refinement. We then allocate individual emission streams to provinces, a 0.05° grid, and administrative districts using process-specific spatial proxies. Uncertainty is quantified using analytical propagation and Monte-Carlo simulation.

The central design principle is that the calculated inventory total serves as the control value. Every provincial, gridded and district product is constrained to reconcile with that control. Emission factors and activity data therefore determine the quantity emitted, while spatial proxies determine only where those emissions are allocated.

1.1. Contributions

This paper makes five main contributions.

First, it provides a reproducible national agricultural inventory implemented in a single codebase that runs on the Python standard library alone. Emission factors, activity values and other parameters are stored in versioned data files rather than embedded in the calculation code. The inventory covers six IPCC categories: enteric fermentation (3.A), manure management (3.B), rice cultivation (3.C), managed soils (3.D), field burning of agricultural residues (3.F), and urea application

(3.H). Liming emissions under category 3.G are not estimated: no national lime-use activity figure has been sought. All six categories use Tier 1 equations with country-specific activity data. Selected livestock parameters, including animal mass and manure-management-system shares, provide partial Tier 2 characterisation. Tier status is therefore reported separately for each category.

Second, the framework allocates emissions to provinces, a national grid and administrative districts under an explicit reconciliation constraint. Within each control area, every emission stream is distributed in proportion to a process-matched proxy and then rescaled so that the allocated values sum exactly to the control total. Livestock emissions are allocated using gridded livestock density and, where available at district level, census head-counts. Rice emissions are allocated using mapped paddy area, while residue-burning emissions are allocated using satellite fire detections.

Third, the inventory is evaluated at several scales. National category totals are compared with FAOSTAT, EDGAR and Climate TRACE using matched years and category definitions. District patterns are evaluated against independent government crop-area records. The rice allocation has a Spearman rank correlation of 0.92 with official district rice area, while the managed-soils allocation reaches 0.84 against district wheat area. Parametric uncertainty is estimated using both IPCC Approach 1 propagation and Monte-Carlo aggregation. These methods apply different propagation formulations to the same underlying uncertainty assumptions.

Fourth, the framework produces a national reconstruction for 1990–2024 and a district-level historical panel based on fixed spatial shares. The national series reflects changes in activity over time, while the district panel distributes each year's national total according to the audited 2024 district pattern.

These are two different products on two different bases and are not interchangeable. The canonical panel is the nine fiscal years FY2015–16 to FY2023–24, computed on Pakistani fiscal activity with the reference year anchored to the 2024 census; it is what the release reports, reconciles and prices uncertainty for. The 1990–2024 series is a contextual reconstruction on FAOSTAT calendar-year activity, provided for long-run trend and for the Common Reporting Table, and it is not the inventory's reporting basis. Where the two overlap they do not agree exactly, and what that change of basis costs is published per year and category in `outputs/pakemis_agriculture_basis_change.csv` rather than being reconciled away.

Fifth, the framework calculates agricultural air-pollutant co-emissions, including NH_3, NO_x and the pollutants released by crop-residue burning. These estimates use the same activity data and process-specific spatial proxies as the greenhouse-gas inventory. A shared, stage-resolved nitrogen ledger aligns the system boundaries used by the IPCC and EMEP/EEA methods and helps diagnose differences between the resulting estimates. The ammonia inventory is also compared with an independent gridded inventory for South Asia. An exploratory Tier-1 land-sector extension is presented separately in Supplementary Note S1 and is excluded from all agricultural results.

Figure 1 summarises the framework. Section 2 reviews relevant inventory and spatial-allocation methods. Section 3 describes the study area and data. Sections 4–6 present the greenhouse-gas inventory, spatial allocation, uncertainty analysis and shared nitrogen accounting. Section 7 describes the evaluation framework. Sections 8 and 9 report the greenhouse-gas and air-pollutant results. Sections 10–12 discuss the findings, limitations and conclusions.

2. Related Work

2.1. National inventory methods

National agricultural inventories generally follow the 2006 IPCC Guidelines (IPCC, 2006) and, increasingly, the 2019 Refinement (IPCC, 2019), which updated several agricultural emission factors and introduced climate-specific disaggregation for soil nitrous oxide.

The IPCC framework defines several methodological tiers. Tier 1 combines default emission factors with national activity data. Tier 2 replaces selected defaults with country-specific parameters or more detailed activity characterisation. Tier 3 uses higher-order models or direct national measurement systems.

Pakistan's national agricultural inventories have been prepared mainly with Tier 1 methods and national activity totals (Global Change Impact Studies Centre, 2019; Mir et al., 2021). This approach supports comprehensive national reporting but provides limited information about subnational patterns or individual processes.

2.2. Global gridded inventories

Several international datasets provide independent estimates of agricultural emissions for Pakistan. FAOSTAT calculates national agricultural and land-use emissions from FAO activity data using mainly IPCC Tier 1 methods and reports results by category and gas (FAO, 2025; Tubiello et al., 2013).

EDGAR provides national totals and gridded emissions at approximately 0.1° resolution using a globally consistent methodology (Crippa et al., 2025, 2020); this paper's comparison table keeps one row from the prior 2024-vintage booklet (Crippa et al., 2024) because the 2025 release revised Pakistan's agricultural CH_4 history and the revision itself is informative. Climate TRACE reports country- and subsector-level agricultural emissions using a combination of inventories, models and observation-informed methods (Climate TRACE, 2026).

These products provide useful external reference points, but they are not substitutes for a nationally compiled inventory. They generally use globally harmonised or model-derived assumptions and may not incorporate the most appropriate national activity data or system definitions. In this study, they are used as external benchmarks rather than as reported national values.

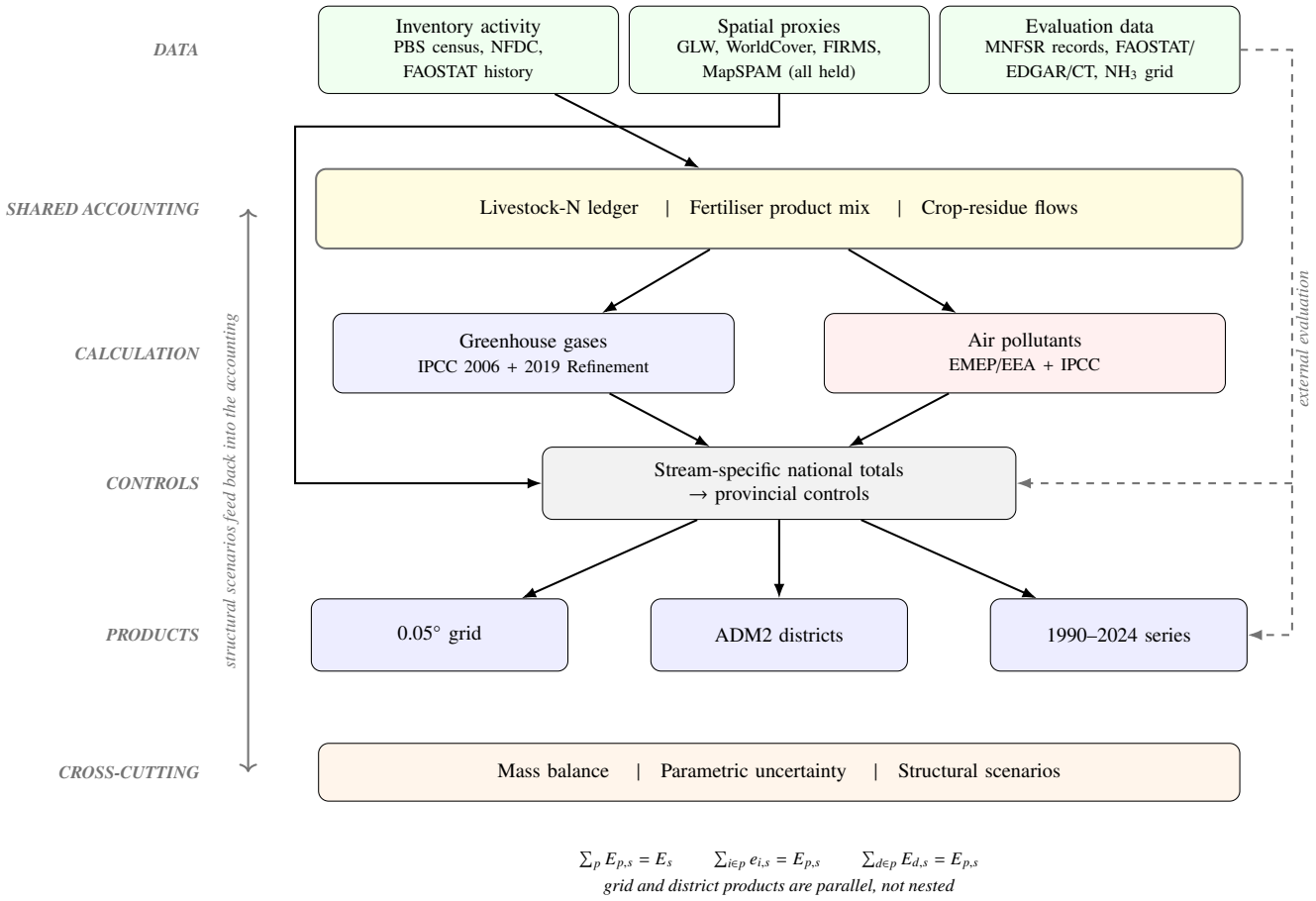


Figure 1: Official activity data feed a shared livestock-nitrogen, fertiliser-nitrogen and crop-residue accounting system. Separate greenhouse-gas (CH_4, N_2O, CO_2) and air-pollutant (NH_3, NO_x, $PM_{2.5}$, CO, NMVOC, BC, SO_2) calculations produce stream-specific national totals, which become provincial controls and are allocated independently to a 0.05° grid and to ADM2 districts on process-matched proxies; every product reconciles to its control. Dashed arrows denote external evaluation, which is applied to the products rather than to the calculations. Evaluation data are never used as inventory inputs.

2.3. *Spatial allocation of emissions*

Allocating national or subnational emissions to a grid is a dasymetric-mapping problem. An emission total is distributed according to a spatial proxy that represents the location or intensity of the underlying activity. Similar approaches are widely used in fossil-CO_2 inventories and global emissions gridmaps (Oda et al., 2018; Crippa et al., 2020).

Several open datasets provide relevant agricultural proxies. The Gridded Livestock of the World (GLW 4) dataset provides modelled species-specific livestock densities (Gilbert et al., 2024, 2018). The Spatial Production Allocation Model (MapSPAM) provides crop-specific physical areas (International Food Policy Research Institute, 2024). ESA WorldCover provides a 10 m land-cover classification (Zanaga et al., 2022), while NASA FIRMS and other satellite products identify active fires and burned areas (NASA, 2025; Giglio et al., 2016).

The quality of a spatial inventory depends on whether each proxy represents the process being allocated. Livestock density is appropriate for emissions produced directly by animals, paddy area for rice methane, cropland for fertiliser-related emissions, and satellite fire occurrence for the location of residue burning. A single proxy applied to an entire inventory category may misallocate emissions when that category contains several distinct activity streams.

A further requirement is mass conservation. Some spatial inventories distribute emissions without subsequently rescaling the mapped values to the original control total. In such cases, the gridded product may no longer reproduce the national or provincial inventory. The framework developed here treats reconciliation as an invariant: every mapped emission stream must sum to its corresponding control total.

2.4. *Agricultural ammonia inventories*

Agricultural ammonia inventories have largely developed within the air-quality community, separately from greenhouse-gas inventories used for climate reporting. The EMEP/EEA Guidebook represents livestock ammonia through a stage-resolved mass flow. Total ammoniacal nitrogen is tracked through grazing, housing, yards, storage and field application, with emission factors varying by animal category and manure-management system (EMEP/EEA, 2023a).

This process resolution is needed because ammonia losses occur at specific stages of manure handling. By contrast, Tier 1

greenhouse-gas methods often represent manure nitrogen using annual fractions that aggregate across several stages. Combining the two frameworks therefore requires careful alignment of their system boundaries and nitrogen pools.

Tomlinson et al. (2026) provide a regional reference for South Asia. Their framework estimates agricultural NH_3 emissions for the SAARC countries across five sectors using an explicit nitrogen-flow model and provides an openly available 0.1° grid (Tomlinson et al., 2025). We use this inventory to compare both the national magnitude and the spatial pattern of our ammonia estimates.

Other global products, including EDGAR and HTAP, also cover Pakistan. However, they are built without national activity data and rely partly on the same underlying FAO statistics used by many national and regional inventories. They therefore provide useful bounds and methodological comparisons, but agreement with them does not constitute fully independent confirmation.

2.5. Crop-residue burning inventories

Crop-residue burning can be estimated using two broad approaches. Bottom-up inventories calculate the mass burned from crop production, residue-to-product ratios, dry-matter fractions, combustion factors, and the proportion of residues burned. National greenhouse-gas inventories generally use this approach, and the burned fraction is its dominant uncertainty.

Top-down or observation-based approaches use satellite fire detections, fire radiative power, or burned-area products. These datasets provide much stronger information about the location and timing of fires, but conversion to burned mass depends on additional assumptions and also carries substantial uncertainty.

This study uses the two types of information for separate purposes. The national quantity burned is calculated using a bottom-up inventory, while satellite fire detections are used only to allocate that quantity spatially. Satellite detections therefore determine the mapped distribution but do not provide an independent estimate of the national total, a distinction that matters because the two are sometimes conflated.

2.6. Contribution relative to previous work

Compared with conventional national inventory workbooks, this framework integrates calculation, uncertainty analysis, spatial allocation, external evaluation and historical reconstruction in a reproducible codebase. Compared with global gridded products, it uses nationally grounded activity data and province-level control totals within a hybrid Tier 1/Tier 2 agricultural inventory.

Its main contribution is the combination of these elements. The national inventory establishes the emission totals, process-specific proxies determine their spatial distribution, and exact reconciliation keeps the provincial, gridded and district products consistent with the reported figures. To our knowledge, this combination has not previously been implemented for Pakistan's agricultural sector.

3. Study Area and Data

3.1. Study area

Agriculture in Pakistan is concentrated in the irrigated Indus basin, which extends through Punjab and Sindh and supports most of the country's crop and livestock production. Approximately four-fifths of Pakistan's cropland is irrigated through canals and tube wells. The country also has a large livestock population, with buffalo accounting for a substantial share, and is among the world's largest consumers of urea fertiliser.

These characteristics shape both the composition and the geography of agricultural emissions. Enteric fermentation and soil nitrous oxide dominate the greenhouse-gas inventory, while emissions are spatially concentrated in the irrigated canal belt.

3.2. Activity data

National activity data are drawn primarily from official Pakistani sources. Livestock populations are taken from the 2024 PBS 7th Agricultural Census (Pakistan Bureau of Statistics, 2025). The Pakistan Economic Survey (Ministry of Finance, 2024) is used as a cross-check. The two sources are not expected to match exactly because the census is based on enumeration, whereas the Economic Survey includes projections derived from earlier census rounds. Within the census itself the national summary table and the district tables are separate transcriptions, and they differ by a handful of head in two species. Neither is edited to force agreement: the national table is the emissions control, the district tables drive the spatial allocation, and the residual per species is published with the normalisation factor it implies in `outputs/agriculture_livestock_reconciliation_2024.csv`.

Crop area and production data are taken from the Pakistan Economic Survey's Agriculture chapter for FY2023–24 (Ministry of Finance, 2024) and PBS agricultural statistics. Synthetic-nitrogen offtake is the audited full-year figure from the National Fertilizer Development Centre's Fertilizer Review for the same year (National Fertilizer Development Centre, 2024). Both documents are archived in this repository; the citations name the editions the engine reads, not the most recent editions published. FAOSTAT activity series are used to reconstruct emissions for 1990–2023, while the 2024 inventory is anchored to the latest Pakistani census and administrative data (FAO, 2025).

3.3. Geospatial data

Spatial allocation uses the open global datasets listed in Table 1.

Livestock emissions are allocated using species-specific densities from Gridded Livestock of the World 4 (Gilbert et al., 2024). Rice-related emissions ride MapSPAM 2020 physical rice area (International Food Policy Research Institute, 2024): the raster (V2r2) is held and hash-pinned since the spatial facility program, its former cropland-fallback class deleted from the allow-list so a checkout without it refuses the build, and the own-year GCD-Rice 2023 classification is published

beside it as a crosscheck and a priced district-level bracket. Crop-residue burning is allocated using NASA FIRMS VIIRS active-fire detections from the reporting period's own burning windows (NASA, 2025). The held record spans all nine fiscal years—1 664 876 unique detections fetched over per-crop burning windows derived from the FAO crop calendar's own harvest dates (rice, wheat, cotton, maize and sugarcane, each harvest followed by a 45-day residue-clearing tail, clipped to its fiscal year)—but the spatial surface consumes only the reporting period's own dates, 166 393 in-grid detections inside FY2023–24. Period-matching is a correction this series already made once (an earlier extraction spanned October 2024–May 2025, one season after the period, and that offset was declared until the period-matched extraction replaced it), so the nine-year record is deliberately not pooled into a climatology here: each year's fires allocate only that year. The window shapes only the within-province fire geography, never a magnitude.

The fire layer's construction is stated in full so its declared choices are reviewable. The product is the science-quality `VIIRS_SNPP_SP` archive, requested in five-day spans (the area API's maximum) over the national bounding box; detections are deduplicated on (latitude, longitude, acquisition date). Each detection is weighted by fire radiative power plus one—FRP approximates biomass burned, and the offset keeps zero-FRP detections from vanishing. Two filters are deliberately *not* applied to this spatial surface, and their absence is a declared choice rather than an omission: no confidence filter (every detection at any confidence enters the surface) and no crop mask (the only mask is the national boundary, so non-agricultural fires inside Pakistan contribute weight where they burn). The monthly timing layer is stricter: there an attribution ladder masks the bounding box's Indian-Punjab detections, excludes persistent detections within 0.01° of the kiln registry's coordinates, and keeps cropland detections only—of the 169 657 detections in FY2023–24, 103 747 fall outside Pakistan, 4102 sit near kilns, 9380 off cropland, and 52 428 are attributed to residue burning; the unattributed counts are published in the burning crosscheck, never forced into the profile. A district shape carrying residue-burning mass in which the layer records no fire falls back to a uniform spread within that shape, declared per event in the grid fallback ledger; a fetch returning zero detections refuses to write at all, so an empty fire layer can never look fetched. Other cropland-related streams are allocated using the cropland fraction from ESA WorldCover 2021 (Zanaga et al., 2022).

Provincial boundaries are obtained from geoBoundaries (Runfola et al., 2020), while district boundaries are based on GADM or GAUL ADM2 polygons. District allocations are evaluated against crop-area statistics published by the Ministry of National Food Security and Research (Ministry of National Food Security and Research, 2023). National emission totals are compared with FAOSTAT, EDGAR and Climate TRACE.

3.4. Status of the estimates

The estimates remain provisional wherever this inventory holds no nationally measured parameter. The livestock population is anchored to the 2024 agricultural census, but the dairy-cattle share is derived from FAOSTAT milk-animal counts. Rice harvested area and several agronomic fractions are based on published literature or default assumptions pending national survey data. Each such parameter is identified in the input files and carried through the resulting outputs and uncertainty analysis.

Dataset	Use	Role
PBS 7th Ag. Census	activity	livestock, crop area
Economic Survey	activity	crops, livestock cross-check
NFDC	activity	synthetic nitrogen
FAOSTAT (QCL/RFN/GT)	activity/bench.	time series, benchmark
EDGAR 2024/2025	benchmark	independent national totals
Climate TRACE v7.1.9	benchmark	independent national totals
MNFSR district crops	records	district validation (rice, wheat)
GLW 4 (2020)	proxy	livestock density
MapSPAM 2020	proxy (held, hash-pinned)	rice area
ESA WorldCover 2021	proxy	cropland fraction
NASA FIRMS (VIIRS)	proxy	fire occurrence
geoBoundaries / GADM	boundary	province / ADM2 district polygons

Table 1: Principal datasets used in the inventory. Pakistani official statistics provide the national and administrative activity data. Open global products provide spatial proxies and external benchmarks. Inputs used only in the exploratory land-sector extension are listed in Supplementary Note S1.

4. Methods

The framework consists of four main stages: calculation of the national agricultural inventory; allocation of national totals to provinces; allocation to a 0.05° grid and to administrative districts; and uncertainty, key-category, completeness and quality-control analysis.

The inventory is predominantly Tier 1. Each category uses the equations in the 2006 IPCC Guidelines, together with relevant updates from the 2019 Refinement, and applies them to country-specific activity data. Selected livestock parameters provide partial Tier 2 characterisation where nationally appropriate values are available. These include typical animal masses, the dairy-cattle share and manure-management-system shares.

Tier status is reported by category in Table 2. This is more informative than assigning one tier to the entire inventory, because the level of country-specific detail differs among categories.

All activity values, emission factors and parameters are stored in version-controlled data files; values are not embedded directly in the calculation code. Greenhouse gases are converted to carbon-dioxide equivalents using AR5 100-year global-warming potentials — 28 for CH_4 and 265 for N_2O — to match the Common Reporting Table convention. AR6-equivalent totals are reported separately.

Category	Method / data	Country-specific elements
Enteric fermentation	T1 method, CS data	species census, dairy split
Manure management	T1/2, CS data	animal mass, Nex, MMS, temp. zone
Managed soils	T1 method, CS data	fertiliser data, dry-climate EFs
Rice cultivation	T1 method, CS data	area, cultivation period
Field burning	T1 method, CS data	crop production, residue params
Urea application	T1 method, CS data	urea offtake; N series before 2024

Table 2: Method-tier status by category. All categories use the Tier-1 (T1) method equations on country-specific (CS) activity data; manure management is partly Tier 2 through country-specific animal mass and management-system shares. No emission factor is Tier 3 (Pakistani field measurement).

4.1. National inventory

Enteric fermentation and manure management. Enteric methane is calculated using the Tier 1 form of IPCC Eq. 10.19,

$$\mathrm{CH}_4^{\mathrm{ent}} = \sum_t N_t \, EF_t \times 10^{-6}, \tag{1}$$

where N_t is the population of livestock category t and EF_t its annual per-head methane emission factor.

Methane from manure management is calculated using IPCC Eq. 10.22 with temperature-dependent factors, and direct nitrous oxide from manure management follows Eq. 10.25. Nitrogen excretion is calculated as $N_{\mathrm{rate},t} \times TAM_t \times 365/1000$, where $N_{\mathrm{rate},t}$ is the daily nitrogen-excretion rate and TAM_t is typical animal mass.

The calculation then distributes excreted nitrogen among manure-management systems. Dung burned as household fuel is assigned to the energy sector and excluded from the agricultural inventory to prevent double counting. Nitrogen deposited directly on pasture is assigned to managed soils rather than to manure management.

Managed soils. Direct soil nitrous oxide follows IPCC Eq. 11.1,

$$\mathrm{N_2O}^{\mathrm{direct}} = \left[(F_{\mathrm{SN}} + F_{\mathrm{ON}} + F_{\mathrm{CR}}) \, EF_1 + F_{\mathrm{PRP}} \, EF_{3,\mathrm{PRP}} \right] \frac{44}{28}, \tag{2}$$

where F_{SN}, F_{ON}, F_{CR} and F_{PRP} are nitrogen inputs from synthetic fertiliser, organic amendments, crop residues and pasture deposition respectively.

Nitrogen applied to rice is assigned the flooded-rice emission factor $EF_{1,\mathrm{FR}}$. Indirect $\mathrm{N_2O}$ from volatilisation and leaching follows IPCC Eqs. 11.9 and 11.10.

For direct $\mathrm{N_2O}$ from non-rice cropland, the main inventory uses the 2019 dry-climate value $EF_1 = 0.005$. This value is appropriate for the arid but irrigated Indus basin and is close to the crop-weighted empirical estimate reported for Pakistan by Cui et al. (2021). The older aggregate value and the wet-climate value are examined in the sensitivity analysis in Section 5.4.

Rice cultivation, residue burning and urea application. Rice methane is calculated using IPCC Eqs. 5.1 and 5.2,

$$\mathrm{CH}_4^{\mathrm{rice}} = \sum_i EF_c \, SF_{w,i} \, SF_{p,i} \, SF_{o,i} \, t_i \, A_i \times 10^{-6}, \tag{3}$$

where EF_c is the baseline daily emission factor, SF_w represents the water regime, SF_p the pre-season water regime, SF_o the effect of organic amendments, t_i the cultivation period and A_i the harvested area.

Emissions from field burning follow the generic biomass-burning equation in the IPCC Guidelines. Urea-application $\mathrm{CO_2}$ follows Eq. 11.13. The urea mass has two bases, and each year states which it used in `outputs/agriculture_urea_basis.csv`. For the reference year it is the national urea offtake NFDC prints for the fiscal year itself. For every earlier year, which has a total-nitrogen figure and no published product split, it is reconstructed from that series by a constant chemical relation. The reconstruction was formerly applied to the reference year as well, where it fell short of the printed offtake by 0.207 %; a measurement is preferred to an approximation of it wherever one exists, and the approximation is retained, and labelled, where none does.

Liming emissions under category 3.G are not estimated: no national lime-use activity figure has been sought. Liming is recorded as "not estimated" in the completeness assessment.

4.2. Provincial allocation

National emission streams are allocated to the four provinces and a small residual region using activity-specific administrative shares.

Enteric fermentation, manure-management emissions and pasture-deposited nitrogen use provincial livestock-census shares. Rice methane and rice-specific fertiliser nitrogen use provincial rice area. Residue-related emissions use cropland shares. Synthetic-fertiliser $\mathrm{N_2O}$ and urea $\mathrm{CO_2}$ use provincial fertiliser-offtake shares.

Applied manure requires a different treatment, because it is generated near livestock but applied to cropland. Its provincial distribution is therefore derived by intersecting cropland with livestock density at the grid-cell level and then aggregating the resulting surface to provinces.

All provincial shares are normalised to sum to one. Provincial estimates therefore reproduce the national control total exactly.

4.3. Gridded allocation and mass-balance reconciliation

Each provincial emission stream is allocated to a 0.05° grid using dasymetric weighting. Let $E_{p,s}$ be the control total for emission stream s in province p, let Ω_p be the set of grid cells within that province, and let $w_{i,s}$ be the spatial-proxy value for cell i. Emissions in each cell are calculated as

$$e_{i,s} = E_{p,s} \, \frac{w_{i,s}}{\sum_{j \in \Omega_p} w_{j,s}}, \qquad i \in \Omega_p. \tag{4}$$

This construction guarantees that $\sum_{i \in \Omega_p} e_{i,s} = E_{p,s}$. The grid therefore reconciles to the provincial total, and provincial totals reconcile to the national inventory.

Spatial proxies are assigned by emission stream rather than by broad inventory category. This distinction matters because a single category can contain several nitrogen or activity streams with different geographies.

GLW 4 livestock density is used for enteric methane, manure-management emissions and pasture-deposited nitrogen. Species densities are weighted by their per-head emission factors, so that the resulting proxy represents the spatial distribution of emissions rather than unweighted animal numbers.

MapSPAM paddy area is the declared proxy for rice methane and the rice-specific fertiliser stream; the raster (2020 V2r2, hash-pinned) is held and consumed since the spatial facility program, its cropland-fallback class deleted fail-closed, and the own-year GCD-Rice 2023 classification ships beside it as a published crosscheck and a priced district-level bracket (Section 3). FIRMS active-fire detections are used for crop-residue burning. WorldCover cropland fraction is used for synthetic fertiliser, crop-residue inputs to soils and urea application.

Managed soils are constructed by allocating each nitrogen input separately and then summing the resulting stream-level surfaces. This prevents the geography of synthetic fertiliser from being imposed on manure-derived or pasture-derived nitrogen.

Applied manure is allocated using cropland weighted by livestock density. This proxy represents two physical constraints: manure is applied on agricultural land, but transport costs generally keep it close to the herd that produces it.

Where a proxy has no positive values within a province, the corresponding emission stream is distributed uniformly across that province. This fallback preserves the provincial control total.

4.4. District allocation and historical district panel

The same proportional-allocation rule is used to distribute provincial totals among administrative districts. The main output contains 116 districts across the four provinces. Gilgit-Baltistan, Azad Jammu and Kashmir, and Islamabad Capital Territory are represented together as a residual unit, producing 115 reporting units in total.

For most emission streams, district shares are obtained by summing the relevant spatial proxy within each district and normalising those sums within the province.

Livestock categories receive a more direct treatment: livestock emissions are distributed using district head-counts from the 2024 PBS Agricultural Census, every census district resolved through the inventory's canonical district crosswalk (several census districts sum onto one boundary polygon where the boundary file predates a split). The two census units the boundary file does not draw, Cholistan Area and Larkana District, keep their own share as named rows of the district panel and fall to the province surface in the grid; a polygon the census does not reach is a build failure, not a proxy.

The gridded and district products are parallel allocations of the same provincial controls. They are not nested representations of one another. The 0.05° grid uses continuous gridded proxies, whereas the district table substitutes census head-counts for livestock where possible. Both products reconcile to the same provincial and national totals, but summing grid cells within a district will not necessarily reproduce the census-controlled district estimate. Formally, $\sum_{i \in \Omega_p} e_{i,s} = E_{p,s}$ for the grid and $\sum_{d \in p} E_{d,s} = E_{p,s}$ for the district product, and both satisfy $\sum_p E_{p,s} = E_s$.

No consistent multi-year district activity dataset is available. The historical district panel therefore holds the audited 2024 district shares fixed and rescales them to each year's national category total for 1990–2024.

The district values consequently follow the national time series. They do not represent observed changes in the geographic distribution of activity over time. This assumption is evaluated in Section 11.

The exploratory land-sector extension in Supplementary Note S1 is excluded from the agricultural totals, benchmarking, key-category assessment and uncertainty analysis.

4.5. Uncertainty analysis

Uncertainty is estimated using two propagation approaches applied to the same set of activity-data and emission-factor uncertainties.

IPCC Approach 1 propagates uncertainty analytically. Uncertainties in multiplicative terms are combined in quadrature, while uncertainties in additive components are combined using their contributions to the total.

The Monte-Carlo analysis samples the *components*: each engine row is drawn from a mean-preserving lognormal at its own declared width—appropriate for non-negative emissions with potentially right-skewed uncertainty—with one shared deviate per declared correlation group, so rows split apart for allocation reasons cannot average their common error away. The simulation uses 10^4 iterations, and the category and sector totals are the *sums of the sampled rows* in every realisation rather than sampled quantities in their own right.

The two approaches test the sensitivity of the reported interval to the form of propagation. They do not provide independent estimates, because both use the same underlying uncertainty assumptions. Agreement between them therefore indicates consistency between analytical quadrature and lognormal aggregation.

Managed-soils components that share an emission factor are treated as perfectly correlated. This applies to separate volatilisation and leaching streams that appear as distinct rows because they use different spatial proxies. Treating them as independent would artificially reduce uncertainty when one physical stream is divided into several reporting components.

Other within-category and cross-category correlations are not modelled *by this sector-scoped stream*, and the qualifier matters because a second engine treats two of them differently. This sector's Monte Carlo samples component totals, which cannot express covariance between the categories

its stages couple; the NATIONAL run instead samples a parameter graph, and there two physical couplings are carried explicitly—fertiliser N as one audited series driving both the soils components and the urea term, and each species' head-count shared between its enteric and manure components. Their effect at national scale is about a hundredth of a percentage point, in the widening direction. Supplementary Note S3 gives that treatment; the limitation of the sector-scoped stream is the one stated here. The reported national interval therefore reflects the documented parameter uncertainties under this correlation structure. Factor-level uncertainties are reported in Table 3, category distributions in Appendix E, and full input provenance in Supplementary Note S3.

4.6. Key categories, completeness and quality control

Key categories are identified using the IPCC Approach 1 level and trend assessments.

The level assessment ranks categories by their contribution to the absolute national total in the reference year. The trend assessment uses $T_x = L_{x,0}\,|g_x - g_{\mathrm{tot}}|$, where $L_{x,0}$ is category x's share in the base year, g_x is its growth since 1990 and g_{tot} is the growth of the total agricultural inventory. The metric therefore identifies categories that are both important in the base year and whose trajectory differs from the sector average. It should not be interpreted simply as a category growth rate.

Categories are accumulated until the 95 % threshold is reached. A category is classified as key if it qualifies under either the level or the trend assessment.

Inventory coverage is documented through a completeness table. Every AFOLU category-and-gas combination receives an IPCC notation key: estimated, not estimated, not occurring, or included elsewhere. This makes known gaps, exclusions and cross-sector transfers explicit.

Quality control is implemented through automated tests covering units, source traceability, mass-balance reconciliation, benchmark ranges, and consistency among national, provincial, grid, district and historical outputs. Quality assurance is supported through comparison with FAOSTAT, EDGAR, Climate TRACE and independent district statistics. Changes to activity data, emission factors or methods trigger recalculation of the complete time series.

5. Emission Factors and Calculated Emissions

The spatial allocation described in Section 4 determines where emissions are placed but does not change their national magnitude. National totals are determined only by the activity data and the emission factors.

Table 3 reports the principal emission factors, their uncertainty, and their IPCC or literature source. Table 4 reports the country-specific and inventory-specific parameters that replace or refine selected Tier 1 defaults. Full source details and derivations are provided in Supplementary Note S2.

Each factor is stored in a version-controlled data file together with its units, source and uncertainty. This structure allows every calculated value to be traced to a documented input.

5.1. Emission factors

Enteric-fermentation and manure-management factors are drawn mainly from the 2006 IPCC defaults for the Indian Subcontinent. Managed-soils factors use the climate-specific values in the 2019 Refinement.

The direct soil N_2O factor is the most consequential individual methodological choice. The main inventory uses the dry-climate value $EF_1 = 0.005$. The older 2006 aggregate value of 0.010 and the 2019 wet-climate value of 0.016 are retained as sensitivity cases.

The dry-climate value is also close to the crop-weighted empirical estimate of approximately 0.0055 reported by Cui et al. (2021) for Pakistan. That empirical value is used as a sensitivity anchor, not as the operative emission factor.

Rice methane uses the IPCC baseline factor together with water-regime scaling. The main case assumes continuously flooded conditions, $SF_w = 1$. Intermittent irrigation and rainfed conditions are evaluated as alternatives.

Factor (unit)	Value	±%	Source
Enteric CH_4 (kg/head/yr; 2006 T10.10–10.11)			
Buffalo	55	40	2006 T10.10
Dairy cattle	58	50	2006 T10.11
Other cattle	27	50	2006 T10.11
Camels	46	50	2006 T10.10
Sheep, goats	5	50	2006 T10.10
Manure management (2006 T10.14–10.21)			
CH_4 buffalo, warm (kg/hd/yr)	5	30	2006 T10.14
CH_4 dairy cattle, warm	6	30	2006 T10.14
N_2O EF_3 solid/liquid (kg N/kg N)	0.005	100	2006 T10.21
N_2O EF_3 drylot	0.02	100	2006 T10.21
Rice CH_4 (2006 T5.11–5.13)			
Baseline EF_c (kg/ha/day)	1.30	53	2006 T5.11
SF_w continuous (default)	1.00	26	2006 T5.12
SF_w intermittent (AWD)	0.52	30	2006 T5.12
SF_w rainfed	0.28	40	2006 T5.12
Managed soils (kg N_2O-N/kg N; 2019 Refinement)			
EF_1 dry (default)	0.005	110	2019 T11.1
EF_1 flooded rice	0.004	120	2019 T11.1
EF_1 aggregate (sens.)	0.010	90	2006 default
EF_1 empirical anchor	0.0055	100	Cui et al. (2021)
EF_4 volatilisation (dry)	0.005	100	2019 T11.3
EF_5 leaching	0.011	100	2019 T11.3
Carbon amendments, burning			
Urea carbon (t C/t urea)	0.20	10	2006 Eq.11.13
Residue burning CH_4 (g/kg d.m.)	2.7	40	2006 T2.5

Table 3: Principal emission and carbon factors, with factor-level uncertainty (±%, 95% relative half-width) and provenance. These factor uncertainties are combined with activity-data uncertainty into the per-category values propagated in the uncertainty analysis (Section 4, Appendix E). Values marked *(sens.)* are used only in the sensitivity analysis (Table 5); the empirical anchor adjudicates the soil-nitrogen choice and is not the operative factor.

5.2. Country-specific and inventory-specific parameters

The largest departures from global Tier 1 defaults occur in the livestock characterisation.

Population-average typical animal masses are used for dairy cattle, other cattle, buffalo and camels—declared judgements resting on working pointers rather than cited measurements, as

Supplement S2's provenance classes record (10 cited, 3 derived, 25 working pointers and 21 declared judgements across the 59 parameter rows). These values are generally higher than the corresponding IPCC defaults and affect both nitrogen excretion and the division of enteric emissions among livestock groups. The dairy-cattle share is derived from FAOSTAT counts of animals reported as being in milk.

Other national parameters determine how fertiliser and soil nitrogen are represented. These include the proportion of synthetic nitrogen supplied as urea, the share applied to rice, and the irrigated fraction used in the leaching calculation.

Table 4 distinguishes among four types of parameter: national activity data, Tier 2 characterisation, regional IPCC defaults, and inventory assumptions. This category-by-category classification defines the methodological basis more accurately than assigning one tier to the whole inventory.

5.3. Calculated emissions

Applying the factors in Table 3 and the parameters in Table 4 to the national activity data produces the 2024 estimates reported in Section 8.

The same emission-factor set is held constant throughout the 1990–2024 reconstruction. Changes in the historical series therefore reflect changes in activity, including livestock growth and increasing fertiliser use, rather than changes in the selected factors.

Appendix C presents the activity data, implied factors and emissions by gas for the 1990 base year and the 2024 reporting year in a Common Reporting Table structure.

5.4. Sensitivity to major methodological choices

Three assumptions account for most of the methodological sensitivity: the direct soil factor EF_1, the rice water-regime factor SF_w, and the distribution of livestock manure among management systems.

Increasing EF_1 from the main-case value of 0.005 to the 2006 aggregate value of 0.010 raises the national total by 11.9 Mt CO_2e. Using the empirical value of 0.0055 produces little change and gives a managed-soils estimate of 29.7 Mt, close to that of Cui et al. (2021).

Changing the rice water regime from continuously flooded to intermittent irrigation reduces the national total by 7.9 Mt CO_2e. A fully rainfed assumption reduces it by 11.9 Mt CO_2e.

Using the GLEAM 3.0 South-Asia manure-management distribution (FAO, 2022) raises the national total by 9.9 Mt CO_2e. Unlike the soil and rice cases, this scenario requires a complete model run because manure management is coupled to the quantity of manure nitrogen later applied to soils. It therefore changes both manure-management and managed-soils emissions.

All tested scenarios remain within the main Monte-Carlo interval and do not change the ranking of the leading emission categories.

6. Shared Nitrogen Accounting and Air-Pollutant Methods

The greenhouse-gas and air-pollutant calculations use a common nitrogen ledger. This ledger tracks how excreted and applied nitrogen moves through manure management, grazing, fertiliser application, volatilisation and soil processes.

The two inventories therefore represent different emissions from the same underlying nitrogen flows. Aligning their system boundaries is necessary to prevent the same nitrogen from being removed twice, retained twice, or assigned inconsistently between NH_3 and N_2O.

6.1. Alignment of the nitrogen boundaries

Two IPCC volatilisation fractions apply at different stages of the manure chain.

$\text{Frac}_{\text{GasMS}}$ (Table 10.22) represents nitrogen lost as NH_3 and NO_x within the manure-management system, principally during housing and storage before manure reaches agricultural land. This loss reduces the quantity of manure nitrogen available for later application.

$\text{Frac}_{\text{GASM}}$ (Table 11.3) represents volatilisation after organic nitrogen has been applied to soil or deposited during grazing. Under the Tier 1 soil-N_2O method this fraction contributes to indirect N_2O. It does *not* reduce the organic nitrogen input before direct soil N_2O is calculated.

Pre-field losses are therefore calculated as

$$\text{Frac}_{\text{LossMS}} = \text{Frac}_{\text{GasMS}} + \text{Frac}_{\text{leachMS}} + \text{Frac}_{\text{N}_2\text{MS}} + EF_3, \quad (5)$$

evaluated separately by livestock species and manure-management system using the factors in IPCC Table 10.22. The post-application factor $\text{Frac}_{\text{GASM}}$ is then applied once, after manure reaches the field or is deposited during grazing.

This implementation replaces an earlier version of the nitrogen chain. Supplementary Note S5 documents the previous errors, their quantitative effects, and the regression tests used to prevent their recurrence.

6.2. Synthetic-fertiliser ammonia

Synthetic-fertiliser NH_3 is calculated from Pakistan's reported fertiliser-product mix rather than from one generic fertiliser composition.

EMEP/EEA factors vary substantially among products because urea and ammonium-based fertilisers have greater volatilisation potential than nitrate-dominant products, particularly on high-pH soils. Pakistan's alkaline cropland and urea-heavy fertiliser mix make product weighting important.

Let w_j be product j's share of applied nitrogen and $EF_j^{NH_3\text{-}N}$ its ammonia-nitrogen emission factor (EMEP/EEA, 2023b). Taking the NFDC offtake by product (National Fertilizer Development Centre, 2024) over one fiscal year, the product-weighted factor is

$$f_{NH_3,SF} = \sum_j w_j \, EF_j^{NH_3\text{-}N} = 0.1608. \quad (6)$$

Parameter	Value (unit)	Status	Source
Dairy (in-milk) cattle share	0.295	national activity datum	FAOSTAT QCL 2024 (16.98/57.54 M)
Typical mass, dairy cattle	400 kg	Tier-2 characterisation	Sahiwal/crossbred; LRRD 2006
Typical mass, other cattle	225 kg	Tier-2 characterisation	Pak. adult zebu population avg.
Typical mass, buffalo	380 kg	Tier-2 characterisation	Nili-Ravi herd avg. (LRRD)
Typical mass, camels	480 kg	Tier-2 characterisation	dromedary herd avg.; Khan 2025
Nex, buffalo	0.32 kg N/1000 kg/d	regional IPCC default	IPCC 2006 T10.19 Asia
Nex, dairy cattle	0.47 kg N/1000 kg/d	regional IPCC default	IPCC 2006 T10.19 Asia
Warm-zone share (manure)	0.70	inventory assumption	Indus-plain climate zone
Irrigated cropland fraction	0.80	national activity datum	FAO AQUASTAT (19/22 Mha)
Rice share of synthetic N	0.16	national activity datum	rice 3.5/22 Mha cropped
Urea share of applied N	0.80	national activity datum	NFDC offtake
Rice cultivation period	125 days	Tier-2 characterisation	PARC calendar (area-weighted)

Table 4: Country-specific and inventory-specific parameters that override IPCC defaults. *Status* distinguishes national activity data, genuine Tier-2 characterisation, regional IPCC defaults, and inventory assumptions — so the paper does not over-claim a uniform Tier-2 basis. Full provenance (source, table/page, derivation, and uncertainty) for these and all other parameters and factors — management-system shares, dung-cake fraction, residue ratios, water/amendment factors, volatilisation and leaching fractions, and the soil N_2O factor set — is given in Supplementary Note S2.

Scenario	Rice	Soils	Total
Main case (EF_1 = 0.005; rice flooded)	16.5	28.5	223.5
EF_1 = 0.010 (2006 aggregate)	16.5	40.4	235.4
EF_1 = 0.016 (2019 wet)	16.5	54.7	249.7
EF_1 = 0.0055 (Cui 2021)	16.5	29.7	224.6
EF_3PRP wet (2019)[b]	16.5	30.3	225.3
EF_3PRP aggregate (2006)	16.5	42.8	237.7
Both 2006 direct factors	16.5	54.7	249.6
Rice intermittent SF_w = 0.52 (AWD)	8.6	28.5	215.5
Rice rainfed SF_w = 0.28	4.6	28.5	211.5
Manure GLEAM 3.0 split[a]	16.5	27.6	233.4

Table 5: Sensitivity of the 2024 national agricultural total (Mt CO_2e, AR5) to the principal methodological choices. [a]Soil and rice alternatives can be recalculated directly because each touches a single category and the affected terms scale linearly. The GLEAM manure-management case requires a complete model run: the nitrogen ledger is coupled, so a different split changes how much manure nitrogen reaches the field and moves managed soils (28.5 to 27.6 Mt) as well as manure management (21.2 to 32.1 Mt). It is the same scenario as Table 23 and reports the same total. [b]The pasture factor had no scenario beside it before this release, which was an asymmetry rather than a judgement: it is the larger lever of the two soils factors—the 2019 Refinement cut the cattle value tenfold from the 2006 default where it halved EF_1—and it carries two-thirds of the difference against FAOSTAT's managed soils. The two factor scenarios are disjoint, so the combined 2006 case is their sum rather than a separate run. Raised via the GLEAM 3.0 South-Asia management-system split (a hidden uncertainty surfaced by the cross-check, not adopted). Generated by `src/pkinv/eval/sensitivity_table.py`.

Urea supplies approximately 81 % of the nitrogen in the product mix and therefore dominates the weighted result (Table 6). The remaining products change the factor by less than 0.01.

The value 0.1608 is already expressed as kilograms of NH_3-N per kilogram of nitrogen applied. It is converted to NH_3 mass by the ratio 17/14 and used directly in the ammonia calculation. It is not multiplied by an additional NH_3 share.

Soil NO_x is calculated separately, using $f_{NO,SF}$ = 0.007 kg NO-N per kilogram of nitrogen applied. For the IPCC indirect-N_2O pathway, which Eq. 11.9 defines on the combined volatilised pool, the two are summed:

$$\mathrm{Frac}_{\mathrm{GASF}} = f_{NH_3,SF} + f_{NO,SF} = 0.1678, \qquad (7)$$

against the IPCC "urea-dominant" default of 0.15. Keeping these three quantities separate preserves the nitrogen balance and avoids applying an additional ammonia fraction to a factor already expressed as NH_3-N. Because $f_{NH_3,SF}$ also drives the greenhouse-gas layer's synthetic indirect N_2O, it is resolved in one place and both layers necessarily use it.

6.3. Livestock ammonia

Livestock NH_3 is calculated using the EMEP/EEA total-ammoniacal-nitrogen (TAN) mass-flow method (EMEP/EEA, 2023a; Sommer et al., 2019). The IPCC post-application fraction $\mathrm{Frac}_{\mathrm{GASM}}$ is not used as a livestock ammonia emission factor, because it was developed for indirect-N_2O accounting rather than stage-resolved NH_3 estimation.

Excreta deposited during grazing lose a species-specific proportion of TAN directly. Manure entering a managed system can lose ammonia during housing or yard exposure, storage and field application. Each stage acts on the TAN remaining after the preceding stage.

The calculation distinguishes among manure systems. Solid storage passes through housing, storage and field-application stages. Daily-spread manure reaches the field without a storage stage. Dry lots are represented as uncovered yards.

Factors also differ among livestock categories, and are taken from Table 3-9 of the Guidebook. Buffalo manure is assigned a TAN fraction of 0.50, compared with 0.61 for cattle, and a storage factor of 0.17 against 0.32. Grazing factors are applied to TAN rather than to total excreted nitrogen (0.14 for cattle and buffalo, 0.09 for small ruminants).

EMEP/EEA does not provide a separate camel category. Camels therefore use the small-ruminant factors as the closest available published analogue, with a wider uncertainty assigned to reflect the approximation.

6.4. Other agricultural air pollutants

Soil NO_x is estimated as approximately 0.7 % of applied synthetic, organic and crop-residue nitrogen, following Stehfest and Bouwman (2006). The value is initially calculated as NO-N and converted to reported NO_x mass using the conventional NO_2-equivalent conversion.

Product	t N	N content	Share of N	$EF^{NH_3\text{-}N}$	Contribution
Urea	3 026 265	0.46	0.8078	0.170	0.1373
DAP	290 777	0.18	0.0776	0.154	0.0120
CAN	206 058	0.26	0.0550	0.043	0.0024
Other NP	202 590	0.20	0.0541	0.154	0.0083
Ammonium sulphate	14 595	0.21	0.0039	0.154	0.0006
NPK	6 162	0.15	0.0016	0.154	0.0003
Total	**3 746 448**		**1.0000**		**0.1608**

Table 6: Derivation of the product-weighted synthetic-fertiliser ammonia factor. The table is generated from `outputs/agriculture_fertiliser_composition_2024.csv`, whose window is exactly the twelve months July 2023 to June 2024; the annexe supplies the normalised composition only, and the nitrogen magnitude remains the audited national offtake. NFDC's national and provincial tables do not sum to each other over the same twelve months: the difference is 15 918 t of product, 0.154 917 % of the national total and mostly urea. It is published by month and product in `outputs/agriculture_nfdc_province_residual.csv` and assigned to Other/unallocated rather than distributed across the provinces, which would make every residual zero and the shares look exact. Product tonnages are NFDC offtake over one fiscal year (National Fertilizer Development Centre, 2024); nitrogen contents and emission factors are EMEP/EEA values for high-pH soils, appropriate to Pakistan's alkaline cropland (EMEP/EEA, 2023b). *All emission factors in this table are kg NH_3-N per kg N applied.* Urea carries 81 % of the nitrogen and therefore dominates the weighting; the audited national nitrogen total (3.775 Mt N) rather than this window's sum is used as the activity datum, with the mix supplying only the composition.

Pollutants from crop-residue burning are calculated from the same dry-matter-burned estimate used in the greenhouse-gas inventory. The burned mass is multiplied by the combustion factor and by pollutant-specific emission factors for CO, $PM_{2.5}$, NMVOC, NO_x, NH_3, black carbon and SO_2 (Table 7). The $PM_{2.5}$ factor is also used by the related satellite-nowcasting layer so that the two products report a consistent crop-burning $PM_{2.5}$ quantity.

All air-pollutant factors are stored with their units, sources and uncertainty in `data/emission_factors.csv`. Their uncertainties are combined with activity-data uncertainty using the same IPCC Approach-1 product rule applied in the greenhouse-gas inventory.

Pollutant / stage	Basis	Factor	±%
NH_3, synthetic fert.	$f_{NH_3,SF}$, kg NH_3-N per kg N	0.1608[b]	30
NH_3, grazing	of TAN; cattle/buffalo | small rum.	0.14 | 0.09	50
NH_3, housing	of TAN; cattle | buffalo | small rum.	0.08 | 0.20 | 0.22	40
NH_3, yard (dry lot)	of TAN; dairy | other cattle	0.30 | 0.53	50
NH_3, storage	of TAN surviving; cattle | buffalo	0.32 | 0.17	40
NH_3, field application	of TAN surviving; cattle | buffalo	0.68 | 0.55	40
TAN fraction	of N excreted; cattle | buffalo | poultry	0.60 | 0.50 | 0.70	20
Soil NO_x	$f_{NO,SF}$, kg NO-N per kg N applied[a]	0.007	100
Crop-residue burning (g/kg dry matter; on $M_{burnt}\cdot C_f$)			
CO	IPCC 2006 T2.5	92	40
$PM_{2.5}$	Akagi et al. (2011)	7.6	50
NMVOC	Akagi et al. (2011) / EMEP	7.0	60
NO_x	IPCC 2006 T2.5	2.5	50
NH_3	Akagi et al. (2011)	1.3	60
BC	Akagi et al. (2011)	0.5	60
SO_2	EMEP/EEA (2019)	0.40	70

Table 7: Air-pollutant factors, calculation basis, and 95% relative uncertainty. Livestock NH_3 follows the species-resolved EMEP/EEA TAN mass-flow method. Crop-burning factors are applied to the same dry matter used in the greenhouse-gas calculation. The synthetic-fertiliser NH_3 factor is already expressed as NH_3-N and receives no additional ammonia-share adjustment. Soil NO_x is calculated as NO-N and reported as NO_2-equivalent mass.

7. Evaluation Framework

The inventory is evaluated through several complementary comparisons. Each addresses a different component of the framework.

First, national and category-level totals are compared with FAOSTAT, EDGAR and Climate TRACE using matched years, gas metrics and category scope. These datasets provide external reference estimates but are not direct observations of national emissions. A ratio between 0.5 and 2.0 is used as a diagnostic screening range. Values outside this range are examined to identify differences in factors, activity data or category definitions. The range is an operational review threshold rather than a formal statistical acceptance criterion.

Second, the model is run for 1990–2023 using FAOSTAT activity series and compared with FAOSTAT emissions. Because the two calculations share much of their activity basis, this comparison primarily tests the implementation and the consequences of different emission factors. It is not an independent test of the underlying activity data.

Third, district spatial patterns are compared with independent government crop statistics that are not used to produce the relevant district allocation. This comparison evaluates the ranking of districts rather than their absolute emissions.

Fourth, reconciliation is tested automatically at every spatial scale. Each grid and district product must reproduce its provincial and national control totals within numerical tolerance. This verifies mass conservation but does not establish the accuracy of the within-province spatial pattern.

Finally, the analytical and Monte-Carlo uncertainty calculations are compared as alternative propagation methods. Since they use the same input assumptions, agreement between them indicates computational consistency rather than independent confirmation.

7.1. District comparison with independent crop records

District allocations are evaluated against district crop-area records published by the Ministry of National Food Security and Research (Ministry of National Food Security and Research, 2023).

Spearman rank correlation is used because the comparison concerns the ordering of districts rather than a claim that crop area directly measures emissions.

The rice-emission ranking is compared with official district rice area. Across 66 matched districts the correlation is $\rho = 0.92$ ($p < 0.001$), and 13 of the 15 highest-ranked rice districts in the inventory are among the 15 highest-ranked districts in the official data.

Managed-soils emissions are compared with district wheat area across 85 matched districts. Wheat is used as a broad indicator of cultivated and fertilised land because it is grown across most cropping districts. The resulting rank correlation is $\rho = 0.84$.

Agreement varies geographically. It is strongest in Punjab and Balochistan and weak in Sindh, where managed-soils emissions follow a rice-and-cotton geography that is not well represented by wheat area alone. The Sindh result therefore reflects a limitation of the comparison variable as well as possible uncertainty in the allocation.

Equivalent independent district checks are not possible for all categories. Livestock emissions already use PBS district census counts where available, so comparison with the same counts would not be independent. No comparable published district dataset was identified for field burning or urea application. These categories are consequently recorded as lacking independent district evaluation rather than being assigned a validation score.

Category	Independent record	n	Spearman ρ
Rice cultivation	District rice area (MNFSR)	66	0.92
Managed soils	District wheat area (MNFSR)	85	0.84
Enteric / manure	— (census is the proxy)	—	n/a
Field burning	— (no district record)	—	n/a
Urea application	— (no district record)	—	n/a

Table 8: Rank comparison between modelled district emissions and independent government crop-area records. Rice emissions are compared with district rice area, and managed-soils emissions with district wheat area. Categories for which the only available district statistic is already used as an input, or for which no suitable district record exists, are reported as not independently evaluated. Generated by `src/pkinv/eval/validation/district_rice_records.py` and `district_records_multi.py`; the same comparison is scored as a registered benchmark task in `outputs/benchmark_spatial_allocation.csv`.

7.2. What each comparison shares with the engine

External is not the same as independent, and a comparison that tuned a product cannot also test it. Table 9 states, for every comparison this section and the release's validation catalogue carry for agriculture, what the comparison shares with the inventory and whether it is a calibration. Three rows carry a weaker claim than they did before this release, and in each case the reason is that the comparison reads something the product is built from.

8. Results

8.1. National greenhouse-gas inventory

Pakistan's agricultural emissions in 2024 are estimated at 223.5 Mt CO_2e using AR5 GWP-100 values. The corresponding AR6 total is 218.1 Mt CO_2e.

Enteric fermentation is the largest category, contributing 151.7 Mt CO_2e, or 68 % of the national agricultural total. Managed soils contribute 28.5 Mt CO_2e, followed by manure management at 21.2 Mt CO_2e and rice cultivation at 16.5 Mt CO_2e. Urea application contributes 4.8 Mt CO_2e, while field burning of crop residues contributes 0.7 Mt CO_2e.

The inventory is therefore dominated by livestock. Enteric fermentation and manure management together account for approximately 78 % of the total. Soil and rice emissions make up most of the remainder.

8.2. Comparison with international inventories

The category-level comparison with FAOSTAT uses 2023, the latest common year, and includes only the five agricultural categories reported by both inventories. Urea and liming are excluded from this comparison because FAOSTAT does not report corresponding lines for Pakistan.

The engine values compared are the calendar-2023 reconstruction of the fiscal series (Table 11), not the FY2023–24 totals reported above, which sit on a different twelve months. For these five categories, our estimate is 207.8 Mt CO_2e, compared with 233.3 Mt CO_2e in FAOSTAT. The resulting ratio is 0.89.

Agreement differs among categories. Enteric fermentation agrees to within five per cent, with a ratio of 0.99. Rice cultivation and field burning have ratios of 1.16 and 0.91 respectively. Manure-management emissions are 1.38 times the FAOSTAT value.

Managed soils show the largest difference. Our estimate is 27.1 Mt CO_2e, compared with 58.5 Mt CO_2e in FAOSTAT, giving a ratio of 0.46. As discussed in Section 10, this difference is driven mainly by the selected direct-soil emission factor: the present inventory uses the 2019 dry-climate factor of 0.005, whereas FAOSTAT retains the older aggregate value of 0.010.

The 1990–2023 reconstruction uses FAOSTAT activity series. The comparison with FAOSTAT therefore primarily evaluates the implementation and the consequences of different emission factors. The close reproduction of the FAOSTAT enteric series indicates that the livestock calculation has been implemented consistently, but it does not independently confirm the underlying activity data.

Full-scope comparisons include urea application and use the GWP basis reported by each benchmark. The inventory is 0.80 times the EDGAR estimate in 2023 and 0.86 times EDGAR in 2024. Climate TRACE reports AR6 values; on that basis, our 2024 total is 0.83 times the Climate TRACE estimate. The higher EDGAR and Climate TRACE totals arise mainly from their larger estimates of enteric methane.

Comparison	What it tests	Shared with the engine	Independence	Calibration
FAOSTAT / EDGAR / CRT categories	Method and emission factors	The national activity statistics both sides read	Partial	No
Cui 2021 direct-N_2O	Direction of the EF1 choice	None stated; the Pakistani value is extrapolated with no in-country observation	Partial	No
District rice area (MNFSR)	Spatial rank of rice emissions	Nothing: the allocation does not read crop area	Large	No
District wheat area (MNFSR)	Spatial rank of managed soils	Nothing	Large	No
Livestock district census	—	The census counts ARE the activity	None	No
NFDC monthly urea offtake	Spread between two monthly shapes	The offtake series IS the adopted profile	None	Yes
VIIRS/MODIS fire detections	Attribution of detections to Pakistani cropland	The detections ARE the burning profile's weights	None	Yes
FAO rice crop calendar	Plausibility of the rice window	Calendar conventions	Partial	No

Table 9: Independence and shared-input matrix for the agriculture comparisons, in the form the aviation paper uses. A comparison whose "shared" column names the product's own input can bound allocation or attribution, and cannot confirm the quantity: the monthly fertiliser and fire rows are calibrations for that reason, and the enteric and manure categories are recorded above as lacking an independent district record rather than being scored against the census that supplies them.

Category (IPCC code)	AR5	AR6	±%
Enteric fermentation (3.A)	151.7	146.2	25
Manure management (3.B)	21.2	21.0	33
Managed soils (3.D)	28.5	29.4	47
Rice cultivation (3.C)	16.5	16.0	61
Urea application (3.H)	4.8	4.8	11
Field burning (3.F)	0.7	0.7	45
Agriculture total	**223.5**	**218.1**	**19**

Table 10: Agricultural greenhouse-gas emissions by IPCC category, 2024, reported in Mt CO_2e using AR5 and AR6 GWP-100 values. The uncertainty column reports the IPCC Approach 1 95 % relative half-width.

Category (2023, AR5)	Engine	FAOSTAT	Ratio
Enteric fermentation	143.2	145.1	0.99
Manure management	20.2	14.6	1.38
Rice cultivation	16.5	14.3	1.16
Managed soils	27.1	58.5	0.46
Field burning	0.7	0.8	0.91
Agriculture subtotal (ex urea/lime)	**207.8**	**233.3**	**0.89**
Full-scope totals (engine incl. urea) vs urea-inclusive inventories			
vs EDGAR 2023 (AR5)	212.6	267.4	0.80
vs EDGAR 2024 (AR5)	223.5	261.2	0.86
vs Climate TRACE 2024 (AR6)[a]	218.1	261.5	0.83

Table 11: Comparison with international agricultural inventories using matched years, category scope and GWP basis. Category-level rows compare the five categories reported by both this inventory and FAOSTAT in 2023. Full-scope rows compare the inventory, including urea application, with EDGAR and Climate TRACE; the EDGAR 2023 row is from the prior 2024-vintage booklet (Crippa et al., 2024) and the EDGAR 2024 row from the 2025 release (Crippa et al., 2025), retained together because the 2025 release revised the agricultural CH_4 history. Climate TRACE is compared using AR6 values.

8.3. Effect of stream-specific spatial allocation

Broad inventory categories are not spatially homogeneous. Managed soils provide the clearest example, because the category includes nitrogen from synthetic fertiliser, applied manure, pasture deposition and crop residues. Approximately three-fifths of the nitrogen in this category is livestock-derived.

Allocating all managed-soils emissions with one fertiliser or cropland proxy would therefore misrepresent much of the category. The framework allocates each nitrogen stream separately using the proxy associated with the process that generates it, and the stream-level surfaces are then summed to produce the category map.

Three allocation changes affect the provincial geography.

First, indirect N_2O from volatilisation and leaching was divided by nitrogen source. Previously each pathway was allocated as one aggregate stream using fertiliser offtake, even though synthetic fertiliser accounts for only approximately 30 % of volatilised nitrogen and 37 % of leached nitrogen. Allocating the synthetic, manure, residue and pasture components separately shifts 2.28 Mt CO_2e out of Punjab.

Second, applied manure was moved from livestock density alone to the cell-level intersection of cropland and livestock density. Livestock density represents where manure is produced, whereas cropland represents where it can be applied; their intersection identifies cultivated areas located close to livestock production. This change shifts 2.08 Mt CO_2e back to Punjab.

The intersection surface is not a simple average of its two components. In Sindh, livestock and cropland are concentrated in the same Indus corridor, so the intersection produces a larger provincial share than either parent proxy alone. In Balochistan, much of the livestock population occupies rangelands with little nearby cropland, so the intersection produces a smaller share than either parent proxy.

Third, synthetic nitrogen applied to rice was moved from provincial fertiliser offtake to mapped paddy area. This produces only a small provincial change.

These revisions redistribute emissions within the fixed national and provincial control system. They do not change the national magnitude. Their combined net provincial effect is small because the first two changes largely offset, but each produces a substantial redistribution when considered separately.

Splitting physical streams also affects uncertainty propagation. Components derived from the same physical flux may use different spatial proxies while retaining a shared emission factor. Treating the components as statistically independent would reduce the uncertainty merely because the stream had been divided into more rows. The analytical calculation therefore com-

bines components sharing EF_4 or EF_5 linearly before applying the quadrature step, and the Monte-Carlo calculation uses one shared deviate for each correlated group. Managed-soils uncertainty remains ±47% under this treatment.

	Change			
Province	**Split indirect**	**Manure proxy**	**Rice proxy**	**Net**
Punjab	−2.28	+2.08	+0.04	−0.17
Sindh	−0.33	+0.50	−0.03	+0.14
Khyber Pakhtunkhwa	+1.17	−1.22	−0.02	−0.08
Balochistan	+1.45	−1.35	+0.01	+0.11

Table 12: Changes in provincial allocation produced by separating indirect N_2O by nitrogen source, replacing the applied-manure proxy, and assigning rice-specific fertiliser to mapped paddy area. Values are reallocations in Mt CO_2e and sum to zero nationally. The small net effect results from offsetting shifts and does not imply that the individual changes are unimportant. Generated by `src/pkinv/eval/proxy_attribution.py`.

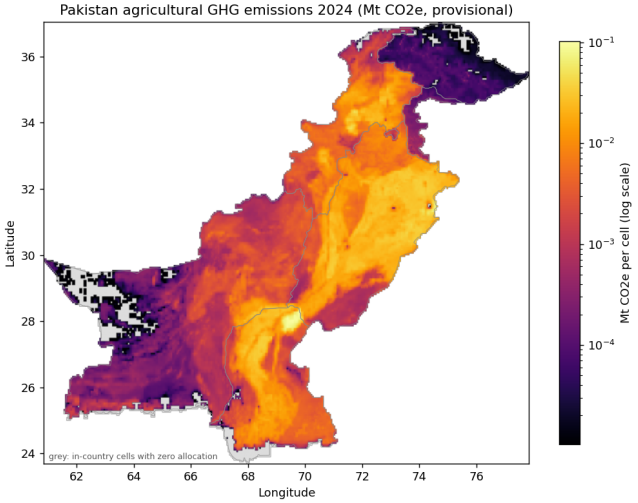


Figure 2: Total agricultural greenhouse-gas emissions on the 0.05° grid in 2024, reported as Mt CO_2e per cell on a logarithmic scale. The gridded surface reproduces the provincial and national control totals.

8.4. Provincial and gridded distribution

Agricultural emissions are concentrated in the irrigated Indus basin.

Punjab contributes 123.3 Mt CO_2e, or 55 % of the national total. Sindh contributes 51.4 Mt CO_2e, or 23 %. Khyber Pakhtunkhwa contributes 31.0 Mt CO_2e, or 14 %, and Balochistan contributes 17.3 Mt CO_2e, or 8 %. A small residual of approximately 0.4 Mt CO_2e occurs outside the four provincial totals.

The 0.05° total-emissions surface follows the canal-irrigated agricultural corridor. Emissions decline sharply across arid western Balochistan and the northern mountain regions.

The category maps have distinct spatial patterns because they use different process-specific proxies. Enteric fermentation and manure management extend into livestock-producing rangelands and settlements. Rice methane follows mapped paddy area. Crop-residue burning is concentrated in the Punjab rice–wheat belt, where satellite fire detections are most frequent.

8.5. District distribution

District allocation resolves the provincial totals to 116 districts and one residual reporting unit (114 with a boundary polygon, and the two census units—Cholistan Area and Larkana District—that the boundary file does not draw, carried as named rows on the province surface)—the four-province base that matches the provincial scope of the activity statistics, narrower by design than the 134-district census frame the residential and waste sectors allocate over. This scale corresponds more closely to the administrative level at which agricultural and air-quality interventions are commonly implemented.

Most of the highest-emitting districts are located in the central and southern Punjab canal belt, with one exception at the head of the table. Karachi has the largest estimated total at 10.18 Mt CO_2e: the 2024 census enumerates the division's seven districts separately—2.8 million cattle and 3.6 million buffalo, Malir's peri-urban dairy colonies the largest part—and all seven resolve onto the one Karachi polygon through the canonical district crosswalk, where earlier releases had placed the shape on the GLW surface (0.10 Mt) because none of the seven census names matched it. Jhang follows at 8.61 Mt CO_2e and Sheikhupura at 6.73 Mt CO_2e. Faisalabad and Muzaffargarh contribute 6.3 and 5.8 Mt CO_2e. The rice-belt concentration sharpened in this release: rice cultivation now rides the MapSPAM surface rather than the generic cropland fallback, which lifts the classic rice districts (Sheikhupura, and Sialkot and Kasur into the table) and is scored against the Punjab Crop Reporting Service in the spatial benchmarks.

These districts rank highly because they combine large livestock populations, intensively fertilised cropland and, in some cases, rice production.

Two limits of the district layer are published rather than described, and since rc41 each is priced as the fraction of every affected stream's own mass that rode the substitute surface, at that stream's gas and methane origin—earlier releases quoted the whole module mass of the affected unit, summed across gases without a warming potential. One census unit, Larkana District, has no polygon of its own in the boundary file, so a stream whose district panel places mass in it falls back to a within-province spread: the provincial total is unaffected and the sub-provincial placement of 2.56 Mt CO_2e across the agriculture, residential and waste streams is not made. The same class carries the panel mass of the territories outside the four provinces (ICT, GB, AJK), for which the district frame has no shapes: 0.35 Mt CO_2e across agriculture and residential. In seven shapes the MapSPAM surface shows no paddy at all, so the rice-specific fertiliser stream is spread uniformly inside them rather than by the declared surface, affecting 0.029 Mt CO_2e. All three are listed with their masses and whether they are resolvable in `outputs/pakemis_spatial_exceptions.csv`. Summing the district layer back

Province	Ent.	Man.	Rice	Soils	Urea	Total
Punjab	79.9	10.3	12.2	17.1	3.3	123.3
Sindh	35.8	4.7	3.4	6.4	1.1	51.4
Khyber Pakhtunkhwa	24.3	3.7	0.3	2.5	0.2	31.0
Balochistan	11.4	2.5	0.7	2.6	0.1	17.3
Other	0.3	0.0	0.0	0.0	0.0	0.4
National	**151.7**	**21.2**	**16.5**	**28.5**	**4.8**	**223.5**

Table 13: Provincial greenhouse-gas emissions by category in 2024, reported in Mt CO_2e using AR5 GWP-100. Field burning is included in the total but omitted as a separate column for space. Provincial rows reproduce the national category totals exactly.

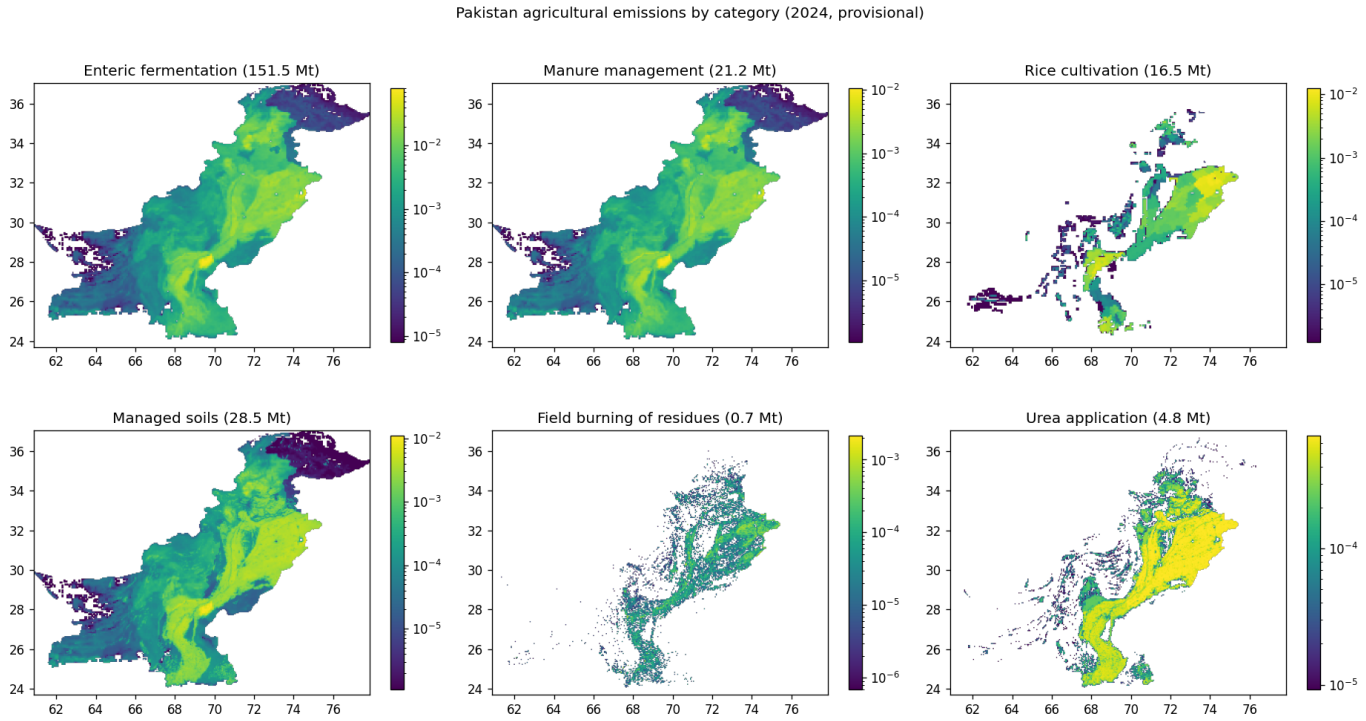


Figure 3: Agricultural emissions by category. Livestock emissions follow species-weighted livestock density, rice methane follows mapped paddy area, and crop-residue burning follows satellite fire occurrence. Each panel uses its own logarithmic scale; colours should not be compared directly between categories.

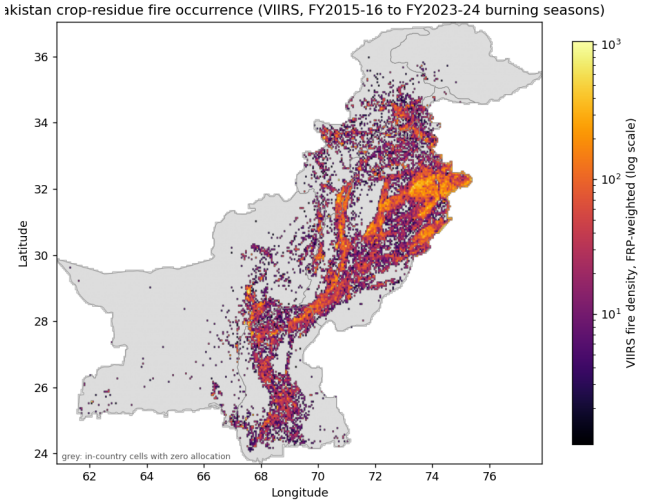


Figure 4: NASA FIRMS VIIRS active-fire detections used to allocate crop-residue burning, from the FY2023–24 reporting period's own season: post-monsoon rice-stubble burning (October–December 2023) and spring wheat-residue burning (April–May 2024), concentrated mainly in Punjab's rice–wheat belt.

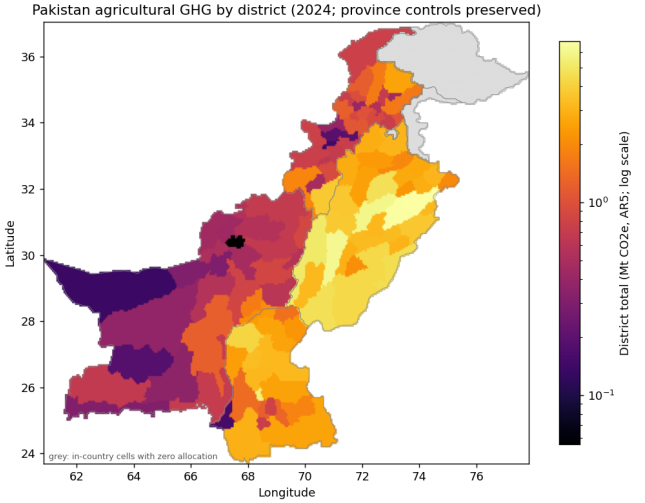


Figure 5: Agricultural greenhouse-gas emissions by district in 2024. Values are shown as annual emissions per unit area, allowing districts of different sizes to be compared. The underlying district totals reproduce the provincial and national inventory.

over each province reproduces the provincial allocation exactly (`outputs/pakemis_adm2_nesting_check.csv`): each shape's mass is partitioned among the reporting units of its own province only, so no mass crosses a provincial boundary between the allocation and the district table.

Matiari in Sindh is the other entry from outside Punjab. Its estimated total is 4.28 Mt CO_2e (ninth). Unlike the leading Punjab districts, its total is driven mainly by livestock density rather than extensive crop cultivation. The category-level decomposition makes this difference visible.

#	District	Province	Mt CO_2e
1	Karachi	Sindh	10.18
2	Jhang	Punjab	8.61
3	Sheikhupura	Punjab	6.73
4	Faisalabad	Punjab	6.32
5	Muzaffargarh	Punjab	5.84
6	Dera Ghazi Khan	Punjab	4.96
7	Bahawalnagar	Punjab	4.47
8	Sargodha	Punjab	4.43
9	Matiari	Sindh	4.28
10	Rahim Yar Khan	Punjab	4.14
11	Okara	Punjab	3.98
12	Bahawalpur	Punjab	3.81
13	Sialkot	Punjab	3.74
14	Kasur	Punjab	3.72
15	Gujranwala	Punjab	3.65

Table 14: Fifteen districts with the highest total agricultural greenhouse-gas emissions in 2024, reported in Mt CO_2e using AR5 GWP-100. District estimates sum to the provincial and national control totals.

8.6. National trajectory, 1990–2024

Agricultural greenhouse-gas emissions increase from approximately 83 Mt CO_2e in 1990 to 223.5 Mt CO_2e in 2024. This corresponds to a compound annual growth rate of approximately 2.9 %.

The increase is driven mainly by growth in livestock populations and synthetic-fertiliser use. Emission factors are held constant throughout the series, so the trajectory represents changes in reported activity rather than changes in methodological assumptions.

The 1990–2023 series uses FAOSTAT activity data on a *calendar-year* basis; the 2024 value is anchored to the PBS agricultural census and other current Pakistani sources on the *FY2023–24 fiscal* basis. The join is therefore a time-basis break as well as a source break—the last FAOSTAT point covers January–December 2023 while the anchor covers July 2023–June 2024, an overlap of six months—and the published series carries a `year_basis` column (`calendar_faostat` / `fiscal_census_anchor`) so no reader can treat the two segments as one homogeneous fiscal series. Part of the change between 2023 and 2024 may reflect either transition. This issue is discussed in Section 11.

8.7. Historical district panel

Applying the fixed 2024 district shares to the national time series produces a district-by-category panel for 1990–2024. The panel contains 24 290 district-year-category records and reconciles to the national inventory in every year.

Under the fixed-share assumption, district trajectories follow the national trend. Sheikhupura increases from approximately 2.9 Mt CO_2e in 1990 to 6.7 Mt CO_2e in 2024. Faisalabad increases from 2.4 to 6.6 Mt CO_2e, and Muzaffargarh from 2.4 to 5.8 Mt CO_2e.

These trajectories should be interpreted as a spatial distribution of the national historical series. They do not show observed district-specific changes in livestock, cropping patterns

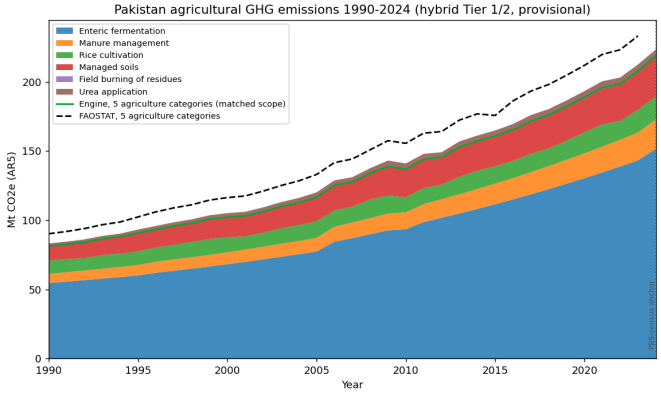


Figure 6: Agricultural greenhouse-gas emissions by category, 1990–2024. The benchmark lines compare this inventory with FAOSTAT using the five categories reported by both systems. The vertical marker identifies the 2024 transition from the calendar-year FAOSTAT activity series to the fiscal-year PBS census anchor—a time-basis break as well as a source break, flagged per row in the published series.

or fertiliser use. Measuring such changes would require consistent historical activity data at district level. The panel now carries that qualification per row rather than only here: every record outside the reference year is marked `reference_shape_backcast`, and only the 694 reference-year records are the audited district pattern. A reader who opens the CSV without this section still cannot mistake a 1990 row for a 1990 observation.

Despite this limitation, the panel provides a district-resolved historical baseline that remains numerically consistent with the national inventory.

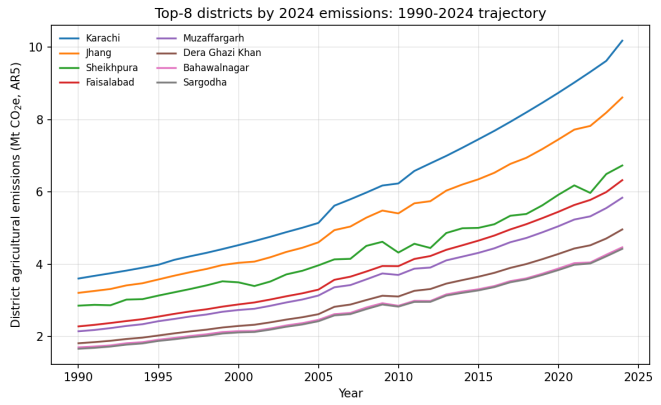


Figure 7: Agricultural greenhouse-gas trajectories for the eight highest-emitting districts, 1990–2024. District values use fixed 2024 spatial shares and reproduce the national control total in every year.

8.8. *Parametric uncertainty*

The IPCC Approach 1 calculation gives a 95 % relative half-width of ±19% for the national agricultural total.

The Monte-Carlo analysis produces a mean of 223.7 Mt CO_2e, close to the deterministic estimate of 223.5 Mt CO_2e. Its median is 222.3 Mt CO_2e, and its 95 % interval is 185.3–270.9 Mt CO_2e. This corresponds to a relative half-width of approximately 19.2 %.

The close agreement indicates that the national interval is not highly sensitive to whether the documented uncertainties are propagated analytically or through mean-preserving lognormal sampling. Both calculations use the same underlying assumptions and the correlation treatment described in Section 4.

Rice cultivation has the widest category-level uncertainty at ±62%, reflecting uncertainty in cultivation conditions and the assumed water regime. Managed soils follow at ±47%, largely because of uncertainty in direct and indirect N_2O factors. Urea application has the narrowest interval at ±11%.

These results show that improved information on rice water management and soil N_2O factors would reduce national inventory uncertainty more than further refinement of the urea calculation.

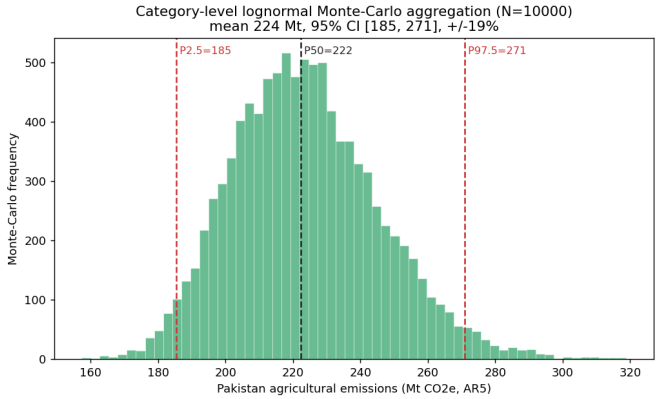


Figure 8: Monte-Carlo distribution of total agricultural greenhouse-gas emissions based on 10 000 draws. The figure marks the median and the 2.5th and 97.5th percentiles.

8.9. *Key categories*

Four categories qualify as key under the IPCC Approach 1 level or trend assessment: enteric fermentation, managed soils, manure management and rice cultivation. Together they account for more than 95 % of both the 2024 inventory level and the contribution to the 1990–2024 trend.

Enteric fermentation dominates the level assessment, contributing 67.9 % of the 2024 total. Rice cultivation contributes only 7.4 % of the 2024 level but accounts for 48.6 % of the trend assessment.

The trend result does not mean that rice emissions grew faster than the other categories. Rice emissions increased by a factor of approximately 1.7 between 1990 and 2024, compared with factors of 2.7–3.2 for the other major categories. Rice ranks highly because its slower growth caused its share of the inventory to change substantially relative to the sector average.

The four key categories include those with the widest uncertainty bands, particularly rice cultivation and managed soils. They are therefore the highest-priority categories for improved activity data and country-specific Tier 2 or Tier 3 emission factors.

Category	Mean	P2.5	P50	P97.5	±%
Enteric fermentation	151.9	117.9	150.7	194.3	25
Manure management	21.2	15.8	20.7	29.8	33
Managed soils	28.5	17.9	27.5	44.5	47
Rice cultivation	16.6	8.7	15.8	28.6	61
Urea application	4.8	4.3	4.8	5.4	11
Field burning	0.7	0.4	0.7	1.0	45
Total	**223.7**	**185.3**	**222.3**	**270.9**	**19**

Table 15: Monte-Carlo means, medians and 95 % intervals for the national agricultural inventory, together with the IPCC Approach 1 95 % relative half-width. Both approaches use the same activity-data and emission-factor uncertainties.

Category	Level %	Trend %	Key
Enteric fermentation	67.9	27.3	✓
Managed soils	12.8	5.0	✓
Manure management	9.5	17.7	✓
Rice cultivation	7.4	48.6	✓
Urea application	2.2	1.2	
Field burning	0.3	0.1	

Table 16: IPCC Approach 1 key-category assessment for agriculture. The level assessment reports each category's share of the 2024 total. The trend assessment measures its contribution to the change in inventory composition between 1990 and 2024. A category is classified as key if it falls within the cumulative 95 % threshold under either assessment.

8.10. Effect of correcting the livestock nitrogen boundaries

The shared nitrogen ledger was used to align the greenhouse-gas and air-pollutant calculations.

Before alignment, the greenhouse-gas calculation appeared to indicate that 21 % of managed nitrogen volatilised, compared with 49.5 % in the air-pollutant calculation. This apparent difference resulted largely from comparing different stages and chemical bases. The IPCC fractions represented both NH_3-N and NO_x-N and separated pre-field losses from post-application losses, whereas the EMEP/EEA calculation represented NH_3-N and followed the complete stage sequence from housing through storage and field application.

After conversion to a common NH_3-N basis, the total livestock volatilisation estimates are relatively close. The IPCC chain gives 1.967 Mt NH_3-N, while the EMEP/EEA chain gives 1.588 Mt NH_3-N. The ratio is 1.24.

The stage-level comparison is less consistent. Housing and storage differ by a factor of 1.25. The IPCC field-application estimate is 0.65 times the EMEP/EEA value, while the IPCC grazing estimate is 3.53 times the EMEP/EEA value. The similar aggregate totals therefore result partly from offsetting stage-level differences.

The greenhouse-gas inventory retains the IPCC nitrogen-accounting chain because the inventory is reported under IPCC methods. The air-pollutant calculation retains the EMEP/EEA stage-resolved method because ammonia losses must be assigned to housing, storage, grazing and field application. Replacing the IPCC loss formulation with the EMEP/EEA formulation in the greenhouse-gas pathway would change the national total by only approximately 0.24 Mt CO_2e. The choice of representation is therefore based mainly on methodological scope rather than on its effect on the headline total.

Correcting the earlier IPCC nitrogen chain reduced the quantity of manure nitrogen reaching soils from 2.72 to 2.33 Mt N. This lowered the agricultural greenhouse-gas total by 0.62 Mt CO_2e, from 221.71 to 221.09 Mt CO_2e. A subsequent update to the synthetic-fertiliser factor produced 221.2 Mt CO_2e; the harmonisation round's activity refresh and the rc39 herd re-base then carried the published total to 223.5 Mt CO_2e. Supplementary Note S5 documents the corrections.

The correction does not change livestock NH_3, because the air-pollutant module reads the managed and grazing nitrogen pools directly rather than deriving ammonia from the final manure-applied pool.

The common ledger guarantees that the same nitrogen is not assigned to multiple stages or omitted between them. It does not require the IPCC and EMEP/EEA methods to produce identical volatilised masses within each stage.

Flow	Mt N	% excreted
A. Excretion partition		
Total N excreted	7.78	100.0
to grazing (F_{PRP})	3.18	40.9
to managed systems	3.34	42.9
diverted as dung-cake fuel (to Energy)	1.26	16.2
B. Pre-field management (Eq. 10.34a)		
NH_3-N + NO_x-N ($Frac_{GasMS}$)	0.91	11.7
leaching, N_2, and N_2O-N	0.09	1.2
= N available for application (F_{ON})	2.33	30.0
C. Field application, gross basis (Ch. 11)		
direct soil N_2O-N (EF_1)	0.01	0.1
NH_3-N + NO_x-N ($Frac_{GASM}$)	0.49	6.3
leaching/run-off	0.45	5.8
= retained in the soil nitrogen pool	1.38	17.8
Same managed pool on the EMEP/EEA TAN basis		
housing, yards and storage NH_3-N	0.69	8.9
field-application NH_3-N	0.71	9.2

Table 17: National livestock nitrogen ledger for 2024. The IPCC chain tracks excreted nitrogen through grazing, managed systems, pre-field losses, field application, leaching and retention in soils. The EMEP/EEA rows provide a parallel TAN-based representation of ammonia losses from the same managed nitrogen pool. Generated by `src/pkinv/sectors/agriculture/n_ledger.py`.

Stage	IPCC	EMEP	Ratio
Housing + storage	0.866	0.694	1.25
Field application	0.466	0.714	**0.65**
Grazing	0.635	0.180	**3.53**
Total	**1.967**	**1.588**	**1.24**

Table 18: Livestock ammonia losses by stage after conversion to a common NH_3-N basis. The IPCC and EMEP/EEA totals are similar, but field-application and grazing estimates differ substantially in opposite directions. Generated by `src/pkinv/sectors/agriculture/n_ledger.py`.

9. Agricultural Air-Pollutant Co-emissions

Agriculture is a major source of ammonia in South Asia. Crop-residue burning also produces particulate matter and combustion gases that contribute to seasonal air-pollution episodes.

The air-pollutant inventory reports emissions in native mass units rather than converting them to CO_2e. The calculation methods and nitrogen accounting are described in Section 6; this section reports the resulting national, spatial, historical and benchmark estimates.

The air-pollutant layer follows the same general processing framework as the greenhouse-gas inventory. Emissions are calculated nationally and allocated to provinces, a 0.05° grid and 116 districts using stream-specific spatial proxies. National time series are reconstructed for 1990–2024, and parametric uncertainty is evaluated analytically and by Monte-Carlo simulation.

The two layers differ in the status of their national totals. The greenhouse-gas total is calculated within the formal IPCC inventory structure. The air-pollutant totals are bottom-up estimates developed in this study. They therefore serve as separate control totals for the air-pollutant spatial products and require external evaluation.

Each air-pollutant grid, district table and provincial allocation reproduces its corresponding national pollutant total to a relative numerical residual below 10^{-9}.

9.1. National air-pollutant totals

Agricultural NH_3 emissions in 2024 are estimated at 2.67 Mt.

Housing, yards and manure storage contribute approximately 0.84 Mt. Manure applied to agricultural soils contributes a further 0.87 Mt. Synthetic fertiliser contributes 0.74 Mt, and grazing deposition contributes approximately 0.22 Mt. Crop-residue burning adds approximately 9 kt.

Crop-residue burning also produces an estimated 654 kt of CO, 54 kt of $PM_{2.5}$, 50 kt of NMVOC, 18 kt of NO_x, 3.6 kt of black carbon and 2.8 kt of SO_2. Agricultural soil processes contribute an additional 145 kt of NO_x, bringing total agricultural NO_x emissions to approximately 163 kt.

Punjab contributes 1.53 Mt of NH_3, followed by Sindh at 0.61 Mt, Khyber Pakhtunkhwa at 0.28 Mt and Balochistan at 0.26 Mt.

The provincial pattern reflects the combination of fertilised cropland and livestock production. Manure-application NH_3 is particularly sensitive to the spatial treatment because it occurs neither strictly where manure is produced nor uniformly across cropland; it is allocated using the intersection of livestock density and cropland.

9.2. Spatial distribution

Each pollutant stream is allocated using the same dasymetric equation applied to the greenhouse-gas inventory.

NH_3 from livestock housing, yards, storage and grazing follows gridded livestock density. At district level, census headcounts replace the gridded proxy where suitable matches are available.

NH_3 from manure application follows cropland weighted by livestock density. Synthetic-fertiliser NH_3 follows cropland, with rice-specific nitrogen allocated to mapped paddy area. Pollutants from residue burning follow satellite active-fire detections.

This common proxy framework keeps the greenhouse-gas and air-pollutant products spatially consistent for nitrogen streams derived from the same activity.

Ammonia emissions are concentrated in the canal-belt and livestock-producing districts. Jhang has the highest district estimate at approximately 93 kt, followed by Muzaffargarh at 85 kt, Sheikhupura at 83 kt, Sargodha at 75 kt and Rahim Yar Khan at 66 kt.

Karachi—the one polygon that receives the division's seven census districts, Malir's peri-urban dairy colonies the largest part—again represents a livestock-dominated profile, whereas the leading Punjab districts combine livestock and intensive cultivation.

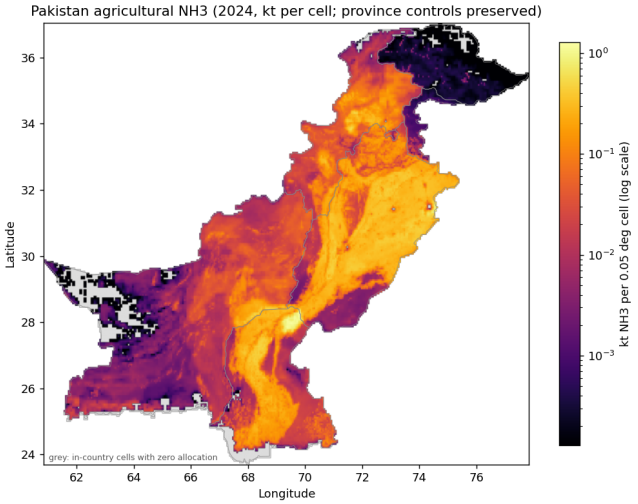


Figure 9: Agricultural NH_3 emissions on the 0.05° grid in 2024, reported as kt NH_3 per cell on a logarithmic scale. Each nitrogen stream is allocated using its process-specific spatial proxy, and the combined surface reproduces the provincial and national air-pollutant totals.

9.3. Historical trajectory

Agricultural NH_3 emissions increase from approximately 784 kt in 1990 to 2675 kt in 2024, a factor of 3.4.

Category (kt)	NH_3	NO_x	$PM_{2.5}$	CO	NMVOC	BC[a]	SO_2
Managed soils	1822	145	—	—	—	—	—
Manure management	843	—	—	—	—	—	—
Field burning	9	18	54	654	50	3.6	2.8
Total	**2675**	**163**	**54**	**654**	**50**	**3.6**	**2.8**

Table 19: Agricultural air-pollutant emissions in 2024, reported in kt of native pollutant mass. Black carbon is included within $PM_{2.5}$ and should not be added to it. NO_x is reported as NO_2-equivalent mass. The air-pollutant values define separate national controls and are not components of the greenhouse-gas CO_2e total. [a]Black carbon is a *component* of $PM_{2.5}$, not additive to it. NO_x is reported as NO_2-equivalent mass. "Managed soils" NH_3 comprises synthetic fertiliser, manure applied to soils and grazing deposition. Uncertainties are given per pollutant in Table 20 and per factor in Table 7. Not part of the GHG control-total reconciliation.

The increase reflects growth in livestock populations and synthetic-fertiliser use. Emission factors remain fixed throughout the series, so changes represent the historical activity data.

CO from crop-residue burning increases from approximately 250 kt to 654 kt, a factor of 2.6. The remaining crop-burning pollutants follow the same underlying change in burned dry matter.

As in the greenhouse-gas reconstruction, the 1990–2023 estimates use FAOSTAT activity data, while 2024 is anchored to the PBS census and current Pakistani statistics. The difference in crop coverage at this transition is discussed in Section 11.

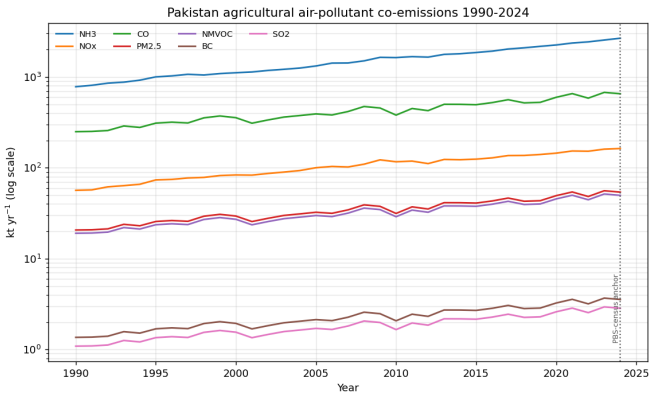


Figure 10: Agricultural air-pollutant emissions, 1990–2024, reported in kt on a logarithmic scale. The vertical marker identifies the 2024 transition from the FAOSTAT activity series to the PBS census anchor.

9.4. *Parametric uncertainty and key sources*

Air-pollutant uncertainty is propagated using the same analytical and Monte-Carlo procedures as the greenhouse-gas inventory.

The Monte-Carlo analysis gives a mean NH_3 estimate of 2676 kt and a 95 % interval of 2120–3359 kt. The corresponding Approach 1 relative half-width is ±24%.

Among the reported pollutants, NH_3 has the narrowest parametric relative interval. Soil and fertiliser NO_x have the widest combined uncertainty at ±90%. Crop-burning uncertainties range from ±54% for CO to ±79% for SO_2.

The comparatively narrow NH_3 parametric interval does not include uncertainty in the national manure-management distribution. That structural uncertainty is evaluated separately in Section 9.7.

Four sources qualify as key for NH_3: manure applied to soils, contributing approximately 32 %; livestock housing and storage, contributing 32 %; synthetic fertiliser, contributing 28 %; and small-ruminant grazing, contributing 4 %. Together these sources exceed the cumulative 95 % threshold. The *ordering* of the three leading sources is soft, and the published ranking now says so quantitatively: under the per-source Monte Carlo their level shares overlap almost completely (each spanning roughly 20–44%), and each holds its point-estimate rank in only about a third to a half of draws (`rank_stability_pct` in `outputs/air_key_categories.csv`). The key-source *set* is stable; which of the three leads is not resolvable at the current uncertainties. Each of the remaining pollutants is dominated by crop-residue burning, so residue burning is its principal key source—at 100 % of the pollutant's total, a statement about coverage (no other stream is estimated) rather than certainty, as the table's note column records.

9.5. *Inventory scope and completeness*

The air-pollutant inventory distinguishes estimated sources, known unestimated sources, and sources outside the agricultural activity basis. Three reporting conventions are important.

First, black carbon is a component of $PM_{2.5}$ and must not be added to the $PM_{2.5}$ estimate. Second, NO_x is reported as NO_2-equivalent mass, a convention that differs from the underlying nitrogen or NO mass used in parts of the calculation. Third, the inventory estimates emissions. It does not estimate ambient concentrations, population exposure or health effects, and no atmospheric transport or chemistry model is applied in this study.

Agricultural machinery and tillage or harvest dust are the main declared source gaps. Suitable activity data and methods for these sources are not included in the present system.

9.6. *Spatial-proxy crosswalk*

Table 22 summarises the spatial proxy used for each emission stream at provincial, gridded and district levels.

Livestock housing and storage use livestock shares at provincial level, GLW 4 density on the grid, and PBS census head-counts at district level. Grazing uses the same broad livestock basis, with species distinctions retained where relevant.

Applied manure uses the cropland–livestock intersection. Synthetic fertiliser uses provincial offtake, cropland on the grid, and district cropland, except for the rice-specific stream, which uses rice area at each scale.

Pollutant	Mean	P2.5	P50	P97.5	±%
NH_3	2676	2120	2657	3359	23
NO_x	163	67	148	349	90
CO	653	376	628	1074	54
$PM_{2.5}$	54	28	52	94	62
NMVOC	50	24	47	93	70
BC	3.5	1.7	3.4	6.6	70
SO_2	2.8	1.2	2.6	5.6	79

Table 20: Monte-Carlo means and 95 % intervals for agricultural air pollutants, together with IPCC Approach 1 relative half-widths. The estimates include parameter and activity-data uncertainty under the reference manure-management configuration but exclude structural uncertainty among alternative management configurations.

Source	NH_3	NO_x	$PM_{2.5}$	PM_{10}	CO	NMVOC	SO_2
Livestock housing and storage	E	NE	NE	NE	—	NE	—
Grazing deposition	E	NE	—	—	—	—	—
Manure application	E	E[a]	—	—	—	—	—
Synthetic fertiliser	E	E	—	—	—	—	—
Residues left in field	E[b]	NE	—	—	—	—	—
Residue burning	E	E	E	NE	E	E	E
Agricultural machinery	NE	NE	NE	NE	NE	NE	NE
Tillage and harvest dust	—	—	NE	NE	—	—	—

Table 21: Completeness assessment for agricultural air pollutants. "Estimated" indicates sources quantified in this inventory, "not estimated" identifies known sources outside the current calculation, and a dash indicates that the source–pollutant combination is not applicable. Black carbon is included within $PM_{2.5}$. [a]Included in the soil NO_x term, which is parameterised on total applied nitrogen rather than per source. [b]Estimated as a computed zero (Section 9.8). Black carbon (estimated for residue burning) is a component of $PM_{2.5}$, not additive to it. Agricultural machinery and tillage dust are outside this inventory's activity basis and are the largest declared gaps.

Crop residues retained in fields use crop production and cropland proxies. Residue-burning pollutants use crop production for provincial controls and FIRMS active-fire detections for grid and district allocation.

Using one crosswalk for the greenhouse-gas and air-pollutant modules prevents corresponding nitrogen streams from being assigned incompatible spatial proxies.

9.7. *Structural uncertainty in manure management*

The ±24% parametric uncertainty for NH_3 holds the assumed distribution of manure-management systems constant. That distribution is itself poorly constrained.

No national survey measures how excreted nitrogen in Pakistan is divided among grazing, confined manure systems and dung used as fuel. Available regional and external parameterisations imply substantially different distributions.

Four configurations were evaluated using the same calculation framework.

The reference configuration is consistent with the FAOSTAT-based manure-management assumptions used in the greenhouse-gas inventory. It assigns 41 % of excreted nitrogen to grazing and produces 2.67 Mt NH_3. Livestock sources account for 1.93 Mt of this total, and the corresponding agricultural greenhouse-gas total under the evaluation harness's own re-implementation is 223.5 Mt CO_2e (Table 23); the inventory's reference total is 223.5 Mt CO_2e (Section 8), and the harness figure is never used as a headline.

A benchmark-like configuration assigns 70 % of excreted nitrogen to grazing. It produces 1.67 Mt NH_3 and a greenhouse-gas total of 217.7 Mt CO_2e.

The GLEAM 3.0 South Asia configuration (FAO, 2022) assigns 31 % to grazing. It produces 3.05 Mt NH_3 and a greenhouse-gas total of 233.4 Mt CO_2e.

A midpoint diagnostic gives 2.46 Mt NH_3 and 224.8 Mt CO_2e. This midpoint is provided only for orientation. It is not evidence-weighted, and the available configurations are not independent observations.

Across the defensible configurations, national NH_3 ranges from 1.67 to 3.05 Mt, and the agricultural greenhouse-gas total varies by approximately 16 Mt CO_2e.

The structural and parametric uncertainties are reported separately. The management configurations are discrete methodological alternatives rather than draws from a known probability distribution, so combining them into one conventional confidence interval would require unsupported assumptions about their likelihood.

The 2.67 Mt value should therefore be interpreted as the reference-configuration estimate. It is used because it keeps the greenhouse-gas and air-pollutant calculations on a common set of management assumptions, not because national evidence demonstrates that this configuration is more probable than the alternatives.

9.8. *Allocation assumptions and residue-derived ammonia*

Manure application. The spatial distribution of manure-application NH_3 is based on a process-informed proxy rather than direct observation.

Livestock density alone would place the emission where manure is generated, including rangelands where little manure is applied to crops. Cropland alone would distribute manure

Emission stream	Province	Grid	District
Housing/storage NH_3, manure CH_4/N_2O	livestock	GLW 4 density	PBS census heads
Grazing NH_3, pasture N_2O	livestock (small rum. split)	GLW 4 density	PBS census heads
Applied-manure NH_3/N_2O	manure-applied	cropland × GLW 4	census × cropland
Synthetic fertiliser, upland	fertiliser offtake	cropland	district cropland
Synthetic fertiliser, rice	rice area	MapSPAM rice (held, hash-pinned)	official rice area
Crop residues (returned)	crop production	cropland	district cropland
Residue burning, all pollutants	crop production	FIRMS active fire	FIRMS active fire
Urea CO_2	fertiliser offtake	cropland	district cropland

Table 22: Spatial proxies used for greenhouse-gas and air-pollutant emission streams at provincial, gridded and district levels. Corresponding nitrogen streams use a shared proxy lookup in both inventory layers.

Management configuration	Grazing share	NH_3 (Mt)	Livestock NH_3 (Mt)	GHG total (Mt CO_2e)
Current (FAOSTAT-consistent)	0.41	2.67	1.93	223.5
Benchmark-like (high grazing)	0.69	1.67	0.92	217.7
GLEAM 3.0 South Asia[a]	0.32	3.05	2.30	233.4
Midpoint diagnostic[b]	0.47	2.46	1.72	224.8

Table 23: Sensitivity of national NH_3 and greenhouse-gas emissions to alternative manure-management distributions. The reference case follows the assumptions used in the main greenhouse-gas inventory. The midpoint is an unweighted diagnostic and should not be interpreted as a preferred estimate. [a]FAO (2022). [b]An unweighted mean of the three configurations, carried as a midpoint for orientation only. It is not evidence-weighted: the three are not independent and no defensible weighting exists, so it should not be read as a best estimate. Generated by `src/pkinv/eval/mms_structural_uncertainty.py`.

independently of nearby livestock supply and transport constraints. The adopted proxy weights cropland by livestock density at the grid-cell level, and therefore identifies cultivated land located close to manure production.

This treatment produces plausible differences among provinces. Sindh receives a relatively high share because intensive cropland and livestock occupy the same Indus corridor. Balochistan receives a lower share because much of its livestock population is located in extensive rangelands with little adjacent cropland.

The proxy does not represent actual hauling distances or manure markets. A national or regional survey of manure transport and application would provide a stronger empirical basis.

Crop residues left in fields. NH_3 from residues left to decompose in fields is estimated as zero rather than omitted.

The adopted EMEP/EEA method produces net NH_3 only when the nitrogen content of the residue exceeds 0.0132. Below this threshold, microbial immobilisation offsets the ammonia that would otherwise be released.

All crops represented in the inventory fall below the threshold. Cotton is closest at 0.0120. Rice is 0.0070, wheat and maize are 0.0060, and sugarcane is 0.0040.

A sensitivity test raises all residue nitrogen contents by 50 %. Under this test cotton exceeds the threshold, but the resulting national emission is only 0.03 kt NH_3. The calculated zero is therefore sensitive to the formal threshold, but the national consequence is negligible.

The external South Asian inventory reports approximately 1.4 kt for this stream using a wider crop set. This difference accounts for only about 0.1 % of its national agricultural NH_3 total and does not explain the broader benchmark difference.

9.9. External spatial comparison

The 0.05° NH_3 surface is compared with the independent 0.1° South Asian agricultural ammonia inventory for 2015 (Tomlinson et al., 2025).

Our inventory is recalculated using 2015 activity and aggregated exactly from 0.05° to the benchmark's 0.1° grid. The comparison covers 8322 grid cells within Pakistan.

Before calculating spatial statistics, the benchmark grid is summed over the national mask. The resulting total is 1052 kt, compared with 1043 kt in the benchmark's published national table. The difference is below 1 %, supporting the grid interpretation, unit conversion and national mask.

At aggregate level the two surfaces show strong spatial agreement: Pearson $r = 0.90$, Spearman $\rho = 0.90$, log-space Pearson $r = 0.74$, and 78 % overlap among top-decile cells.

Despite this spatial agreement, our national 2015 total is 1.76 times the gridded benchmark value. The ratio is similar across Punjab, Sindh and Khyber Pakhtunkhwa, at approximately 1.72–1.77. This indicates that most of the difference is in national magnitude rather than in broad spatial distribution.

Balochistan is an exception, with a ratio of 2.78. Its NH_3 estimate is dominated by extensive livestock systems, making it particularly sensitive to the uncertain division between grazing and managed manure systems.

The aggregate spatial correlation partly reflects shared inputs. Both inventories allocate livestock emissions using GLW 4, and livestock accounts for most agricultural NH_3. The stream-level comparison is therefore more informative.

Synthetic-fertiliser emissions use different underlying crop surfaces in the two inventories. This stream agrees in both pattern and magnitude, with $r = 0.89$, 72 % top-decile overlap and a level ratio of 1.19.

Residue-burning NH_3 also uses different surfaces. Agreement is weak, with $r = 0.38$, 39 % top-decile overlap and a level ratio of 0.21.

The comparison therefore supports the fertiliser allocation most strongly. It shows that the livestock difference is mainly one of magnitude, while the burning stream differs in both pattern and level.

Two qualifications remain. The inventory uses fixed-vintage within-province spatial proxies, so changes between 2015 and 2024 arise from provincial controls rather than from evolving internal patterns. In addition, the sector definitions of the two inventories are similar but not identical.

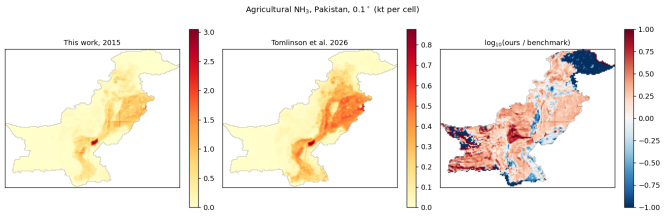


Figure 11: Agricultural NH_3 emissions over Pakistan in 2015 at 0.1° resolution: this inventory, the independent South Asian benchmark, and their ratio. The two surfaces show strong spatial agreement but differ in magnitude, with the largest relative difference over Balochistan's rangelands. Generated by `src/pkinv/eval/spatial_nh3_validation.py`.

9.10. Comparison of national ammonia estimates

National NH_3 is compared with the South Asian benchmark at the benchmark year of 2015.

Matching the comparison year is important. Agricultural NH_3 in the present reconstruction increases by a factor of 1.44 between 2015 and 2024. Comparing the benchmark's 2015 value directly with the 2024 anchor would combine methodological differences with nine years of activity growth.

For 2015 this inventory estimates 1862 kt NH_3, compared with 1043 kt in the South Asian benchmark (Tomlinson et al., 2026). The ratio is 1.79. Comparing the same benchmark with the 2024 estimate would produce a ratio of 2.56. The 2015 estimate is 1.48 times the corresponding EDGAR v8 value and 1.36 times the HTAP v3 value.

The uncertainty intervals provide a stronger diagnostic than the operational factor-of-two screening range. The South Asian benchmark reports a 95 % interval of approximately 764–1322 kt for Pakistan. The present inventory's 2015 interval is approximately 1423–2307 kt. The intervals do not overlap, and the uncertainty-normalised difference is $z = 3.1$. The national difference is therefore larger than the parametric uncertainty reported by either inventory.

Agreement varies sharply by source. Synthetic-fertiliser NH_3 is 1.20 times the benchmark value. Livestock NH_3 is approximately 2.50 times the benchmark. Residue-burning NH_3 is 0.22 times the benchmark value.

The shared nitrogen ledger helps diagnose the livestock difference. After conversion to common boundaries and an NH_3-N basis, the IPCC and EMEP/EEA livestock chains produce totals that differ by only a factor of 1.24. However, both remain substantially above the external benchmark.

This suggests that the difference cannot be attributed solely to choosing the IPCC or EMEP/EEA loss formulation. Their stage-level estimates remain different, so the formulation may still contribute, but the shared excreted-nitrogen pool and the assumed manure-management distribution also require examination.

Country-specific animal masses increase estimated nitrogen excretion. Buffalo are assigned a typical mass of 380 kg, compared with the IPCC default of 295 kg, while dairy cattle are assigned 400 kg rather than 275 kg. The greenhouse-gas inventory shows a related signal: manure-management emissions are 1.38 times the FAOSTAT value.

The external benchmark does not report an implicit emission factor for buffalo, despite buffalo being a major component of Pakistan's livestock population. A residual calculation based on its published national total and the species it reports implies a buffalo factor no greater than approximately 3.9 kg NH_3 per head per year, as set out in Supplementary Note S4.

The present reference configuration gives approximately 9.8 kg per buffalo per year, and the GLEAM 3.0 configuration gives 11.7 kg per head per year. These external parameterisations therefore imply markedly different assumptions about the treatment of buffalo manure, particularly the proportion deposited during grazing.

The inferred benchmark factor is not a direct observation. It is calculated as a residual and also absorbs unresolved contributions from camels. It should therefore be treated as an indication of where the discrepancy may arise rather than as a definitive attribution.

The available evidence points most strongly to the manure-management distribution, with animal-mass assumptions and stage-specific loss formulations as additional contributors. Because the manure-management distribution has not been measured nationally, the livestock NH_3 estimate remains provisional.

The lower residue-burning estimate reflects the wider set of burned crops included in the external inventory. Its additional 1.4 kt estimate for residues left in fields is too small to explain the national difference and would increase rather than reduce the gap.

Stream	Our spatial input	r	Hotspot	Ratio	Shared?
Synthetic fertiliser	WorldCover (upland; rice on MapSPAM)	0.89	72%	1.19	no
Livestock (all)	GLW 4	0.91	78%	2.47	**yes**
Residue burning	FIRMS active fire	0.38	39%	0.21	no
All streams	—	0.90	78%	1.76	partly

Table 24: Stream-level spatial comparison with the independent South Asian agricultural NH_3 inventory for 2015. The livestock comparison uses a shared GLW 4 spatial product and is therefore not fully independent. Fertiliser and residue-burning comparisons use different spatial inputs in the two inventories. Generated by `src/pkinv/eval/spatial_nh3_validation.py`.

			Matched year (2015)		Anchor (2024)	
Scope	**Benchmark**	**Bench.**	**Engine**	**Ratio**	**Engine**	**Ratio**
Synthetic fertiliser	SANH 2026	514	617	1.20	737	1.43
Livestock (all streams)	SANH 2026	496	1238	2.50†	1928	3.89†
Residue burning	SANH 2026	32	7.0	0.22†	9.2	0.29†
National total	SANH 2026	1043	1862	**1.79**	2674	2.56†
National total	EDGAR v8[a]	1257	1862	**1.48**	2674	2.13†
National total	HTAP v3[a]	1372	1862	**1.36**	2674	1.95

Table 25: Agricultural NH_3 compared with external inventories at the benchmark year of 2015 and, for context, at the 2024 anchor. Matched-year comparisons separate methodological differences from activity growth. Ratios outside the study's diagnostic screening range are marked for further investigation. [a]Implied from the percentage differences reported by Tomlinson et al. (2026). One scope mismatch is not corrected for: four NH_3 streams against the study's five sectors. Generated by `src/pkinv/sectors/agriculture/air_pollutants.py`.

10. Discussion

10.1. The managed-soils difference is methodological

The engine's managed-soils estimate is about half of FAOSTAT's. The difference primarily reflects the selected direct emission factor. The 2019 Refinement's dry-climate $EF_1 = 0.005$ was derived largely from *irrigated* dry-climate fields, making it the best-supported value for the irrigated Indus belt; the wet-climate value is not supported there, and the empirical crop-weighted estimate of Cui et al. (2021) for Pakistan (≈ 0.0055) is close to it. FAOSTAT still uses the older 2006 aggregate default $EF_1 = 0.010$. At that aggregate value the engine's managed soils rises to 40.4 Mt CO_2e and the national total to 235.4 (Table 5), closing most of the gap. We therefore report the disaggregated value as the default and the aggregate value as a sensitivity, rather than tuning to match the benchmark.

10.2. Effect of country-specific livestock parameters

The principal country-specific refinements occur in livestock characterisation rather than soil factors: the country animal live weights that set nitrogen excretion (particularly for buffalo) and the FAOSTAT-derived dairy split. This indicates where survey effort would most improve the inventory — livestock characterisation and manure-management shares — the categories that also carry the Tier-2 parameters in Table 2.

10.3. Value of the reconciliation invariant

Enforcing Eq. 4 on every run is what makes the maps usable alongside the reported totals. It also decouples the two design choices that are often conflated: activity data and emission factors determine the national magnitude, while the spatial proxies determine only its geographic distribution and cannot alter the national total. This separation makes the system auditable — a reviewer can check the totals and the proxies independently.

10.4. What the district resolution adds

Resolving to districts changes what the inventory can support. The category decomposition distinguishes two district profiles that a single activity proxy would blur: the crop-and-livestock canal-belt districts of central Punjab, and a livestock-dominated outlier such as Karachi, whose ranking comes from the peri-urban dairy colonies of Malir rather than cropping. This is the same lesson the source-specific-proxy literature reaches from the opposite direction — that where different sources dominate, one generic proxy misplaces emissions — and it is why the livestock split is put on real census head-counts rather than a gridded surface. The independent crop-record check (Section 7.1) then tells us where the district geography is trustworthy: strong for rice and, via wheat area, for managed soils in Punjab and Balochistan; weak in Sindh, where the wheat footprint is the wrong reference; and simply unvalidated for the categories with no independent district statistic. Reporting that unevenness, rather than a single headline correlation, is what keeps the district layer honest.

10.5. Implications for inventories in data-scarce settings

The approach generalises beyond Pakistan. Combining coarse but reliable national/provincial control totals with independent, stream-specific spatial proxies — and substituting a dedicated enumeration (here, the livestock census) for the one category where a generic proxy is weakest — addresses a structural problem common to most low- and middle-income inventories, where activity data are published only at a coarse administrative resolution. The reconciliation invariant keeps

the resulting maps mass-conserving and consistent with the reported totals, so spatial detail cannot alter the reported national figure. Where the method still falls short — the Tier-1/2 factor basis, the constant multi-year district shares, and the categories lacking an independent district record — is exactly where country-specific measurement, not more sophisticated disaggregation, would have the largest effect.

11. Limitations

The estimates are provisional and desk-based. Several limitations follow directly from the method and should condition interpretation of the results.

Emission factors and tier status

1. **Tier 1/2, not Tier 3.** No emission factor is backed by Pakistani field measurement; the factors are IPCC defaults elevated with country-specific activity data. The widest uncertainty bands — rice and soil nitrous oxide — reflect this, and it is the principal limitation on inventory confidence.
2. **Emission-factor context.** With the exception of the empirical soil-nitrogen anchor, factors originate from IPCC regional defaults; combustion, manure, and soil processes in Pakistani conditions may differ systematically from the values underlying those defaults, in either direction.
3. **The soil direct factor is a deliberate choice.** We use the 2019 dry-climate $EF_1 = 0.005$, which halves managed soils relative to the 2006 aggregate 0.010 that FAOSTAT retains (Section 10). We report the aggregate value as a sensitivity (~40 Mt) rather than tune to the benchmark; the choice is defensible but consequential.
4. **Rice is modelled as continuously flooded** ($SF_w = 1$). The intermittent and rainfed water-regime scaling factors exist in the factor set but are not applied, so alternate-wetting-and-drying mitigation is not represented — a real simplification on a sizeable, policy-relevant category with a wide uncertainty band.

Livestock and manure management

5. **Manure-management shares are uncertain.** The management-system split is Tier-2/FAOSTAT-consistent; an alternative South-Asian split would move manure direct nitrous oxide materially, and the split is not measured for Pakistan.
6. **The dung-cake transfer has an observed destination for part of its mass.** Dung-cake combustion (~28% of manure) is removed from this sector's manure-management balance. The national reference books as household fuel what the household survey observes, under a fifth of the diverted mass; the remainder is carried as an unresolved row outside the total with no emission applied, and its candidate fates are priced rather than chosen. Under national configuration L, the reference of earlier releases and now the alternative, all of the diverted mass is RECEIVED by the residential sector as household fuel, and there the cross-module closure is enforced with an exact equality, not merely declared (an earlier edition of this item still called the receiving account out of scope).
7. **The crop-residue fuel transfer is closed the same way, with a larger open remainder.** Of 86.14 Mt of dry residue, 9.27 Mt (122 192 TJ) is diverted to household fuel and received by the residential sector's fuel ledger under the same exact-equality discipline as dung cake; 20.40 Mt is returned to soil and 8.37 Mt burned in the field. The 48.10 Mt the ledger cannot place (55.8 % of dry residue—fodder, bedding, industrial use and decay) is carried as an explicit unresolved row rather than assigned a fate.
8. **Livestock ammonia is ~3× independent inventories, and is provisional.** Benchmarking (Section 9.10) shows our synthetic-fertiliser NH_3 at 1.20× and the livestock streams at 2.5× an independent South Asian inventory (Tomlinson et al., 2026), both compared at the benchmark's own year. The shared nitrogen ledger (Section 8.10) makes the loss formulation a secondary rather than a primary suspect, but does not eliminate it: the IPCC and EMEP livestock totals agree to within 1.23× on a common basis, yet field application and grazing disagree by factors of 0.66 and 3.5 in opposite directions and partly offset. What it does *not* do is establish which side is right, because both representations act on the same excreted-nitrogen pool — raised by our Tier-2 animal masses, which also leave manure management at 1.38× FAOSTAT — and the benchmark's own livestock parameters are not fully reproducible (its implicit-emission-factor table omits buffalo, the dominant species here). We therefore treat the livestock NH_3 estimate and the manure-derived N_2O that shares its nitrogen as provisional. Matching the benchmark's *year* removed roughly a third of the apparent gap, and back-calculating its unpublished buffalo factor (Section 9.10) points most of the remainder towards the manure-management split rather than the animal masses or the loss chain. That split is not measured for Pakistan, and the two external parameterisations that bear on it — GLEAM 3.0 and this benchmark — bracket our assumption from opposite sides and disagree with each other by a factor of three. A national manure-management survey is therefore the single highest-value data collection identified by this work, and it would move both the NH_3 and the manure CO_2e estimates.

Historical and spatial assumptions

9. **Historical source and time-basis transition.** The 1990–2023 series uses FAOSTAT activity data on a calendar-year basis while 2024 is anchored to the PBS census on the FY2023–24 fiscal basis; no statistical harmonisation is applied across this join—neither for the source change nor for the six-month basis offset—so part of the 2023–2024 change may reflect either transition rather than true annual activity growth (livestock species are mapped to a common six-species taxonomy, but levels can differ). Crop coverage also differs: the historical file

carries four crops where the 2024 anchor carries five, so residue-driven quantities — the burning greenhouse gases and the burning air pollutants — are a few per cent lower before the join than after it.

10. **Gridded and district maps are allocated surfaces, not observations.** Within each province the pattern follows a proxy; the reconciliation invariant guarantees the totals but not the sub-provincial detail.

11. **District spatial shares are held constant across years.** The district × year panel (Section 8.7) rescales a single 2024 spatial pattern to each year's national total, because no multi-year district activity survey exists. It distributes national growth across space; it does not detect district-specific shifts in activity.

Validation and uncertainty

12. **District validation is uneven.** Only rice and (via wheat area) managed soils have an independent district record; the livestock, burning and urea-application categories have none, and their district rankings are therefore unvalidated against official statistics (Section 7.1).

13. **Uncertainty treats most components as independent.** Perfect correlation is modelled among terms that share an emission factor within managed soils — the indirect volatilisation and leaching streams, which are separate rows only because they follow different spatial proxies. All other correlations (shared activity data, common factor families across categories) are omitted *here*, because this stream samples component totals rather than parameters, and could widen or narrow the national interval. The national engine samples a parameter graph instead and carries two of these couplings explicitly; Supplementary Note S3 gives their magnitude.

14. **The reported ammonia interval is parametric, not total.** The Monte-Carlo band holds the manure-management configuration fixed and varies factors and activity within it. But that configuration is itself the dominant uncertainty: across defensible alternatives national NH_3 spans 1.67–2.67 Mt (Section 9.7), a wider range than the ±25% parametric band. The two are reported separately and should not be combined into a single interval, because the structural term is a choice between discrete configurations rather than a distribution.

15. **Air-pollutant totals are bottom-up, not control totals.** The air-pollutant layer now carries the same gridded, district, time-series, Monte-Carlo and key-source treatment as the greenhouse gases, and its spatial products reconcile to its national totals by construction. But those national totals are our own bottom-up estimates rather than an audited IPCC control total, so what the allocation conserves is weaker in provenance than for the GHGs. The crop-burning factors are IPCC/EMEP/biomass-burning defaults, soil NO_x is a small high-uncertainty term, and none of the layer is field-validated here.

Scope and omitted sources

16. **Not-estimated agriculture gaps.** Soil-organic-nitrogen mineralisation (F_{SOM}) and liming are marked NE in the completeness table (Appendix D); they require field or national activity data. (Land-sector gaps — SOC change, harvested wood products — are covered in Supplementary Note S1.)

17. **Land sector excluded.** The exploratory Tier-1 land sector is the least settled part of this work (single-year, contested mangrove sign) and is reported only in Supplementary Note S1, excluded from all agricultural results.

18. **The fire layer is an observation, and observations can be absent.** The burning surface now rides the reporting period's own detections (the offset window that stood here—October 2024–May 2025, one season late—was replaced by the period-matched extraction). The honest cost is sparsity: a district shape carrying residue-burning panel mass in which VIIRS recorded no fire falls back to a uniform spread within that shape, declared per event in the grid fallback ledger—where the dense cropland proxy it replaced could never say a district had no observed fires.

Together these bound the inventory as a transparent, reproducible *backbone* rather than a measured account: its magnitudes are auditable and its geography reconciles by construction, but Tier-3 factors, a resolved rice water regime, and district-level ground data — each identified above — are where further effort would most improve it.

12. Conclusion

We have presented a reproducible system that computes agricultural greenhouse gases and air pollutants for Pakistan from one shared activity and nitrogen accounting, and distributes both to provinces, a 0.05° grid and 116 districts under a reconciliation invariant. For 2024 it estimates 223 Mt CO_2e and, in native mass, 2.7 Mt NH_3 with 0.65 Mt CO and 54 kt $PM_{2.5}$ from crop-residue burning. The ammonia figure is the estimate under the manure-management configuration the greenhouse-gas inventory also uses; across defensible alternative configurations it spans 1.67–2.67 Mt, and that structural range, rather than the parametric interval, is the honest measure of how well the national total is constrained.

Two results seem to us to generalise beyond Pakistan.

The first is that *agreement in aggregate does not imply agreement in the components*, and this recurred at two scales. Putting the IPCC and EMEP/EEA nitrogen chains on common boundaries made their livestock totals differ by only 1.24×, but no individual stage agreed: field application and grazing disagreed by factors of roughly two and three in opposite directions and partly cancelled. The same pattern appeared in the spatial allocation, where two large proxy errors had offset almost exactly. An inventory that checks only its totals will not see either.

The second is that *spatial pattern and emission magnitude can be separately right or wrong*. Compared against an independent gridded ammonia inventory at its own year, our surface

agrees well on pattern while sitting 1.8 times higher in level, by a near-constant factor across provinces. Reported per stream, and setting aside the livestock comparison where both inventories distribute on the same livestock product, the synthetic-fertiliser stream agrees on pattern *and* level, while the burning stream agrees on neither. A single national ratio would have hidden all of this, and so would a single correlation coefficient.

What remains open is specific rather than diffuse. The manure-management partition is not measured for Pakistan, the two external parameterisations that bear on it disagree by a factor of three, and it dominates both the ammonia estimate and the manure nitrous oxide. A national manure-management survey is the single highest-value data collection this work points to. Beyond it, the burning geography needs an evaluation against something other than the fire product used to allocate it. A monthly layer for the sector now ships with the national release (`outputs/agriculture_monthly_2024.csv`, and as this sector's slice of the nine-year `outputs/pakemis_monthly_fy2015_16_fy2023_24.csv`), and by the release's own declaration—84 % of the sector's emissions ride a flat, declared profile. That figure is true and, read alone, misleading: most of the flatness is correct rather than missing, because enteric fermentation and manure management are continuous processes riding a census stock and no monthly acquisition could improve a flat month there. Separating the two readings (`outputs/agriculture_timing_classes.csv`) leaves 6.67 % of the nine-year panel, and 7.00 % of the reference year, whose timing this inventory should resolve and has not—almost all of it managed soils. That, rather than the flat share, is the remaining temporal work. An exploratory land-sector extension is provided only as Supplementary Note S1 and is excluded from all agricultural results.

The design choice underlying all of it — treating the national inventory as a control value to which every spatial product must reconcile, while keeping emission factors and spatial proxies as separable, auditable components — resolves the usual tension between spatial detail and reported figures. The result is a backbone a national team can run, audit and update from a documented pipeline, and one that states plainly which of its numbers are settled and which are not.

That design generalises past agriculture. The same control-total discipline now carries nine further sectors on the same engine, and the reconciliation it enforces within this sector is enforced between sectors too: the dung-cake nitrogen this inventory removes from the manure-management balance is accounted for on the receiving side through the same herd and the same volatile-solids rates—the part the household survey observes as household fuel in the residential sector, the remainder as an unresolved row outside the total—and the crop residue diverted to household fuel (9.27 Mt dry, 122 192 TJ) is received by the same fuel ledger from the residue balance, so neither pair can drift apart (Ahmad, 2026). Agriculture is 50.2 % of that national total and the largest contributor to its uncertainty, which is why the manure-management survey named above is a national priority and not only an agricultural one.

Code and data availability

The full system — the calculation engine for both the greenhouse-gas and air-pollutant layers, the shared nitrogen ledger, the gridding and district pipelines, the versioned data files, the test suite and the outputs reported here — is version-controlled in the inventory repository (https://github.com/RehanAhmad13/pakistan-emissions-sectors). It reproduces from a single command (`run_all.py`) with NO user-supplied input: the FIRMS extraction is committed and hash-pinned, and the MapSPAM rice raster is held and hash-pinned beside it (data/gis/manual/RICE_SOURCES.json) with its cropland-fallback class deleted from the allow-list, so a checkout without the raster refuses the build rather than silently substituting (Appendix A). DOIs are minted at archival and none exists yet. Activity data derive from public sources listed in Section 3; downloaded caches are regenerated by the pipeline. The independent ammonia grid used for spatial validation is redistributed by its authors under an open licence (Tomlinson et al., 2025).

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Appendix A. Reproducibility

The inventory runs from a documented pipeline (`python3 run_all.py`) that executes the agricultural stages — national inventory, provincial downscaling, gridding, district downscaling, time series, Monte-Carlo, key-category, sensitivity, and uncertainty-input analysis, completeness table, and a consolidated dashboard — and writes all tables and figures reported here. The exploratory land sector is an *optional* module (skipped under `-quick`; see Supplementary Note S1) and feeds none of the agricultural results. The agricultural core runs on the Python standard library alone; the gridding and figures use an optional geospatial stack. The shipped surfaces need nothing supplied: the FIRMS extraction and the MapSPAM rice raster are both committed-or-held and hash-pinned, so "one command" means one command (the default contract is set out below). Every emission factor, parameter, and activity value is stored in a versioned data file, and a suite of more than 240 unit tests checks the IPCC worked-example calculations, exact reconciliation at each scale (national, provincial, gridded, district, and the district × year panel), emission-factor traceability, and the key-category, sensitivity, uncertainty, and completeness outputs. The suite also covers the air-pollutant engine and its shared nitrogen ledger — ledger closure at every stage, the separation of the pre-field and post-application volatilisation fractions, conservation between the NH_3-N and NH_3+NO_x bases, the province and grid allocation of every pollutant, the matched-year and spatial benchmarks, and the structural manure-management scenarios — together with a consistency check that every headline number and figure in this manuscript still matches the output file it was generated from.

Code and data availability.. The full source code, data files and tests are version-controlled; DOIs are minted at archival (none exists yet, and no identifier is printed before it resolves), and the archived release corresponding to any submitted version will let reviewers run the same code that produced these numbers. The archive contains the calculation engine for both layers, the shared nitrogen ledger (`outputs/n_ledger_2024.csv`, every stage-flow row as kilograms of nitrogen and as a share of what was excreted) and its diagnostic runs (`outputs/n_ledger_stage_comparison.csv`, the IPCC and EMEP ammonia-nitrogen factors set against each other stage by stage), the versioned emission-factor and activity files, the gridded (GeoTIFF, EPSG:4326) and district (CSV) products, the 1990–2024 time series, the uncertainty and benchmark outputs—including `outputs/air_uncertainty_montecarlo.csv`, the per-pollutant intervals with the Approach 1 comparison beside them—and the test suite; units, coordinate reference system and time coverage are recorded with each product, and a run manifest records the exact inputs behind the tables reported here. Reproduction: install the (optional) geospatial stack and run `python3 run_all.py`. The shipped surfaces reproduce as-is with no user-supplied input; the contract is stated in full in Appendix B. Expected outputs are the CSV tables and PNG figures reproduced here (written under `outputs/`). The deposited capsule carries its own single command, `sh capsule/run_tests.sh`, which reruns the agriculture suite in a fresh interpreter with nothing installed; in the full repository `python3 -m pytest tests` runs the test suite, as does `python3 -m unittest discover tests`, which is what continuous integration uses. On the released commit both pass in full; on any other commit the release-integrity tests fail by design: they compare the committed release artefacts—manifest, cover, checksums, PDFs—against the working tree, and they can only pass on the exact commit the artefacts were cut from. They are not a failed reproduction, and a reader who sees them should not read them as one. The code is released under the MIT licence and the derived data under CC-BY-4.0; third-party inputs retain their original licences.

Appendix B. Data acquisition

THE REPRODUCTION CONTRACT, stated once. Livestock density (GLW 4, 2020) is retrieved directly from the FAO open cloud store. The FIRMS VIIRS extraction is *committed* and hash-pinned in `data/gis/manual/`, so no access key is needed to reproduce (a key is needed only to RE-FETCH, which would create a new extraction). MapSPAM 2020 is *held* (V2r2, hash-pinned in `RICE_SOURCES.json`) and its former cropland-fallback class is deleted from the allow-list, so a checkout missing the raster refuses the build rather than silently substituting. The DEFAULT reproducible surface is therefore: MapSPAM for rice, frozen FIRMS for burning, and nothing to supply. Re-fetching FIRMS or swapping the rice surface (the own-year GCD-Rice 2023 classification ships as the priced alternative) changes the surface and is an *upgrade*: it creates a new configuration and a new manifest, and its outputs are not the ones reported here. The pipeline auto-detects supplied files and logs which proxy each category used.

Appendix C. IPCC common reporting table

Table 26 gives a Common-Reporting-Table view of the agriculture sector for the base year (1990) and the latest year (2024): the underlying activity data, the implied emission factor, and the resulting CO_2e. The principal emission and carbon factors themselves are listed in the body (Table 3).

Code / category	Gas	Activity (2024)	1990	2024
3.A Enteric ferment.	CH_4	245.52 M ruminant head	54.5	151.7
3.B Manure mgmt.	CH_4	245.52 M ruminant head[a]	4.3	13.2
3.B Manure mgmt.	N_2O	7.78 M t N excreted[a]	2.4	8.0
3.C Rice cultivation	CH_4	3.64 M ha paddy	9.6	16.5
3.D Managed soils	N_2O	F_{SN} 3.78, F_{ON} 2.34, F_{PRP} 3.18, F_{CR} 0.21 M t N[d]	10.3	28.5
3.F Field burning	CH_4,N_2O	7.11 M t d.m. combusted[c]	0.3	0.7
3.H Urea application	CO_2	6.58 M t urea[b]	1.9	4.8
Agriculture total			**83.2**	**223.5**

Table 26: IPCC CRT-style summary by category and gas, 1990 and 2024 (CO_2e in Mt, AR5). [a]Manure CH_4 is driven by the ruminant herd and its N_2O by the nitrogen excreted by that herd; both are independent of the synthetic-fertiliser series. Poultry adds to the head-count but little emission. [b]Urea mass is NFDC's printed national offtake for the fiscal year; earlier years, which have no published product split, keep the derivation from the national synthetic-N series, and each year states which it used. [c]The residue activity is dry matter actually *combusted*: production × residue ratio × burned fraction gives 8.4 Mt of dry matter reaching the fire, of which the combustion factor $C_f = 0.85$ leaves 7.1 Mt burnt. Emission factors are applied to the latter. The full destination accounting is published in `outputs/agriculture_crop_residue_ledger_2024.csv`. Liming (3.G): not estimated, assumed negligible on Pakistan's alkaline, calcareous soils pending national lime-use data. [d]The four nitrogen inputs of Eq. 11.1, from the shared ledger: synthetic fertiliser, applied organic manure, pasture deposition and crop residue. They are not summed here because the pasture term carries its own factor (EF_{3PRP}).

Appendix D. Completeness and QA/QC

Table 27 gives the notation-key coverage of the agriculture categories. All major agriculture source categories are represented, with the not-estimated (NE) gaps shown explicitly: soil-organic nitrogen mineralisation (F_{SOM}), which requires field data; and liming, set to NE (assumed negligible on Pakistan's alkaline, calcareous soils pending national lime-use data) rather than assumed not to occur. Dung-cake combustion is the one cross-sector transfer to Energy. Land-sector completeness (forest, cropland/grassland SOC, wetlands, harvested wood products) is documented in Supplementary Note S1. Air-pollutant co-emissions (NH_3, NO_x, $PM_{2.5}$, CO, NMVOC, BC) are estimated for the relevant categories in Section 9 and reported in native mass, outside this CO_2e completeness table; agricultural SO_2 is estimated but small (~2.6 kt, from residue burning only). Quality control is automated (source traceability, unit checks, exact reconciliation, benchmark bands); quality assurance is the independent FAOSTAT and EDGAR comparison; and any method or factor change — such as the harmonisation of the volatilisation factor EF_4 to the dry-climate value across the manure and soils terms — triggers a full 1990–2024 recalculation to preserve time-series consistency.

Category	Gas	Key
3.A Enteric fermentation	CH_4	E
3.B Manure management	CH_4, N_2O	E
3.C Rice cultivation	CH_4	E
3.D Agricultural soils	N_2O	E
SOC mineralisation (F_{SOM})	N_2O	NE
3.E Savanna burning	—	NO
3.F Field burning of residues	CH_4, N_2O	E
3.G Liming	CO_2	NE
3.H Urea application	CO_2	E
Dung-cake combustion	multi	IE

Table 27: Completeness and notation keys for the agriculture sector (IPCC Vol. 1 Ch. 8). E estimated, NE not estimated, NO not occurring, IE included elsewhere (Energy). Land-sector categories are in Supplementary Note S1.

Appendix E. Uncertainty inputs

The uncertainty analysis (Section 4) is reproducible from the inputs below. Each category's combined uncertainty is a two-stage quadrature: every emission component combines an activity-data and an emission-factor uncertainty (Eq. 3.1), and the components are aggregated to the category (Eq. 3.2). The full underlying-input table — per component, the activity-data uncertainty, the factor uncertainty, their sources, and which inputs are shared across categories — is given in Supplementary Note S3; the factor uncertainties also appear as the ±% column of Table 3. Table 28 gives the resulting per-category distribution used by the ANALYTIC comparison alone—the Monte-Carlo samples the engine's components, not these fitted category distributions: the point estimate E_c, the combined 95% half-width, and the mean-preserving lognormal parameters $\sigma_{\ln} = (\text{half-width}/100)/1.96$ and $\mu_{\ln} = \ln E_c - \sigma_{\ln}^2/2$. Correlation is modelled in one specific place and not elsewhere, and the distinction matters: components that are separate rows only because they follow different spatial proxies — the managed-soils indirect volatilisation and leaching streams — share a single emission factor (EF_4 and EF_5 respectively), so they are treated as perfectly correlated, summed linearly before the quadrature step and given one shared deviate per group in the Monte-Carlo. Treating them as independent would shrink the reported band purely because the rows were cut more finely. All other within- and cross-category correlations (shared activity data, common factor families across categories) are *not* modelled as shared deviates *in this sector-scoped stream*, which samples component totals; the national run's parameter graph does carry two of them, and Supplementary Note S3 reports their effect. Approach 1 combines these analytically; the Monte-Carlo samples the *components* themselves—each row drawn mean-preserving lognormal at its declared width,

with the correlated soils groups receiving one shared deviate per draw—and the category and sector totals are the sums of those sampled rows in every realisation.

Category	E_c **(Mt)**	**±% (95%)**	$\mu_{\ln}$	$\sigma_{\ln}$
Enteric fermentation	151.7	25.4	5.013	0.129
Manure management	21.2	32.9	3.042	0.167
Managed soils	28.5	47.1	3.323	0.237
Rice cultivation	16.6	60.9	2.760	0.304
Field burning	0.7	44.9	–0.426	0.226
Urea application	4.8	11.2	1.571	0.057
Total	**223.4**	**19.1**	—	—

Table 28: Derived per-category uncertainty parameters. $\mu_{\ln}, \sigma_{\ln}$ are the mean-preserving lognormal parameters implied by each category's 95% half-width, and they describe an ANALYTIC comparison, not the simulation: the Monte-Carlo samples the engine's *components* (with shared deviates inside declared correlation groups) and obtains each category as the sum of its sampled rows. The national total is likewise a sum, never a sampled input distribution, so no $\mu_{\ln}/\sigma_{\ln}$ is given for it. Underlying activity/factor inputs and sources are in Supplementary Note S3. Generated by `src/pkinv/eval/uncertainty_inputs.py`.